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MIL-STD-2301
18 June 1993

DEPARTMENT OF DEFENSE INTERFACE STANDARD

COMPUTER GRAPHICS METAFILE (CGM) IMPLEMENTATION STANDARD FOR THE NATIONAL IMAGERY TRANSMISSION FORMAT STANDARD

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FOREWORD

1. The National Imagery Transmission Format Standards (NITFS) is the standard for formatting digital imagery and imagery-related products and exchanging them among members of the Intelligence Community (IC), as defined by Executive Order 12333, the Department of Defense (DOD), and other departments and agencies of the United States Government as governed by Memoranda of Agreement (MOA) with those departments and agencies.

2. The National Imagery Transmission Format Standards Technical Board (NTB) developed this standard based upon currently available technical information.

3. The DOD and members of the Intelligence Community are committed to interoperability of systems used for formatting, transmitting, receiving, and processing imagery and imagery-related information. This standard describes the Computer Graphics Metafile (CGM) implementation and establishes its application within the NITFS.

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Defense Information Systems Agency (DISA), Joint Interoperability and Engineering Organization (JIEO), Center for Standards (CFS), Attn: TBCF, 11440 Isaac Newton Square, North, Reston, VA 22090 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard establishes the requirements necessary to implement Computer Graphics Metafiles (CGM) used for the representation of symbol graphics in the National Imagery Transmission Format Standard (NITFS).

1.2 Content. This standard provides technical detail of CGM commands, formats, and implementation used for the CGM implementation for NITFS.

1.3 Applicability. This standard is applicable to the Intelligence Community and the Department of Defense. This standard defines the subset of commands (correlated with the minimum implementation subset of commands specified in MIL-D-28003) applicable for graphic annotation of imagery within the NITFS. It is mandatory for all Secondary Imagery Dissemination Systems (SIDS) in accordance with the memorandum by the Assistant Secretary of Defense for Command, Control, Communications, and Intelligence, ASD(C^3I), Subject: National Imagery Transmission Format Standard (NITFS), 12 August 1991. This directive shall be implemented in accordance with the Joint Interoperability and Engineering Organization (JIEO) Circular 9008 and MIL-HDBK-1300. New equipment and systems, those undergoing major modification, or those capable of rehabilitation shall conform to this standard.

1.4 Tailoring task, method, or requirement specifications. The minimum compliance requirements for implementation of CGM are defined in Defense Information Systems Agency (DISA)/JIEO Circular 9008.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the DOD Index of Specifications and Standards (DODISS) and supplement thereto, cited in this solicitation.

STANDARDS

FEDERAL

FED-STD-1037B	-	Telecommunications: Glossary of Telecommunication Terms, 3 June 1991.

FEDERAL INFORMATION PROCESSING STANDARDS (FIPS)

FIPS PUB 128	-	Computer Graphics Metafile (CGM) [adaptation of American National Standards Institute/International Organization for Standardization (ANSI/ISO) 8632:1992].

MILITARY

MIL-STD-2500	-	National Imagery Transmission Format for the National Imagery Transmission Format Standards (NITFS), 18 June 1993.
MIL-D-28003A	-	Military Representation for Communication of Illustration Data: CGM Application Profile, 15 November 1991.

HANDBOOK

MIL-HDBK-1300	-	National Imagery Transmission Format Standards (NITFS), 18 June 1993.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

(Copies of Federal Information Processing Standards (FIPS) are available to DOD activities from the Standardization Documents Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094. Others must request copies of FIPS from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161-2171.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified. Unless otherwise specified, the issues are those cited in the solicitation.

DISA/JIEO Circular 9008	-	NITFS Certification Test and Evaluation Program Plan, (Effectivity 8).

(Copies of DISA/JIEO Circular 9008 may be obtained from DISA/JIEO/JITC/TCBD, Fort Huachuca, AZ 85613-7020.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified. Unless otherwise specified, the issues of the documents adopted by the DOD are those listed in the issue of the DODISS cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI X3.4 - 1986	-	American National Standard Code for Information Interchange (ASCII), 1986

Application for copies should be addressed to the American national Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited, the text of this standard shall take precedence. However, nothing in this standard shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The following definitions are applicable for the purpose of this standard. In addition, terms used in this standard and defined in the FED-STD-1037B shall use the FED-STD-1037B definition unless noted.

a.	ANSI	American National Standards Institute
b.	ASCII	American Standard Code for Information Interchange
c.	ASD(C^3I)	Assistant Secretary of Defense for Command, Control, Communications and Intelligence
d.	CGM	Computer Graphics Metafile
e.	DISA	Defense Information Systems Agency
f.	DOD	Department of Defense
g.	DODISS	Department of Defense Index of Specifications and Standards
h.	FIPS	Federal Information Processing Standard
i.	IC	Intelligence Community
j.	ISO	International Organization for Standardization
k.	JIEO	Joint Interoperability and Engineering Organization
l.	LSB	Least Significant Bit
m.	MOA	Memoranda of Agreement
n.	MSB	Most Significant Bit
o.	NITF	National Imagery Transmission Format
p.	NITFS	National Imagery Transmission Format Standards
q.	NTB	National Imagery Transmission Format Standards Technical Board
r.	RGB	Red, Green, Blue
s.	SIDS	Secondary Imagery Dissemination System
t.	VDC	Virtual Device Coordinates

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3.2 Definitions used in this standard. The definitions used in this document are defined as follows:

a. Character - 1. A letter, digit, or other symbol that is used as part of the organization, control, or representation of data. 2. One of the units of an alphabet. Note: For MIL-STD-2301, a character (ANSI 3.4-1986 7-bit ASCII code padded into 8-bits) is an unsigned integer between and including 32 and 126 and is specified in this document using the character array C1, C2, ... Cn.

b. Commands - For MIL-STD-2301, commands are CGM statements that denote a state to act upon when CGM is read sequentially. The words "command" and "element" are used synonymously throughout MIL-STD-2301.

c. Computer Graphics Metafile (CGM) - CGM is a set of basic elements for a computer graphics data interface usable by many graphics-producing systems and applications.

d. Elements - For MIL-STD-2301, elements are CGM statements that denote a state to act upon when the CGM is sequentially read. The words "command" and "element" are synonymously used in MIL-STD-2301.

e. Integer parameters - For MIL-STD-2301, all integer parameters are 16-bit two's complement signed integers except where specified. Each 16-bit word is numbered from most significant bit to least significant bit using 15 to zero (as illustrated on figure 1). When a 16-bit two's complement integer is used as a parameter in a CGM metafile, the high order byte of the integer is represented as the 8 most significant bits. That is, bits 15 through 8. Bits 7 through zero represent the low order byte of the integer. Note: This is also known as the "Big-Endian" or "Network Byte Order" representation for 16-bit integers.

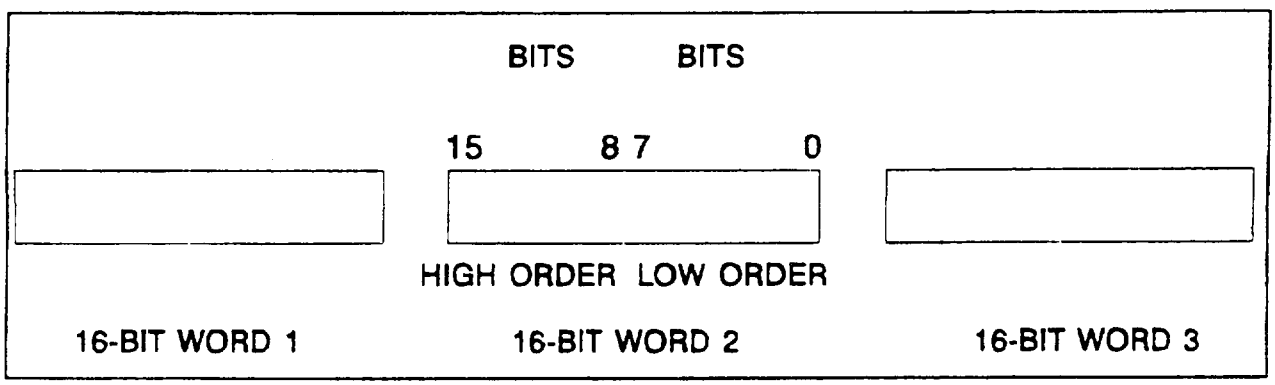


FIGURE 1. Integer parameters.

f. Virtual Device Coordinates (VDC) space - The VDC space defines a coordinate system that is overlaid onto an image to which CGM elements are referenced.

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4. GENERAL REQUIREMENTS

4.1 CGM commands. The CGM implementation for NITFS is a subset of the Federal Information Processing Standard FIPS PUB 128 and complies with the specifications established in FIPS PUB 128. The CGM structure is explained fully in the FIPS PUB 128 document. The following describes the required CGM commands that the CGM implementation for NITFS must support for input interpretation and output generation and are grouped by element class. All CGM commands used in this subset are listed by element class along with their encoding in the CGM Interface Input Requirements and CGM Interface Output Requirements sections of this document. When CGM commands are encountered that are beyond the minimally compliant CGM implementation listed below, the interpreter is responsible for interpreting, discarding, or substituting for these commands.

a. Metafile Delimiter Elements
   - BEGIN METAFILE
   - BEGIN PICTURE
   - BEGIN PICTURE BODY
   - END PICTURE
   - END METAFILE

b. Metafile Descriptor Elements
   - METAFILE VERSION (version = 1)
   - METAFILE ELEMENT LIST
   - METAFILE DESCRIPTION
   - FONT LIST

c. Metafile Picture Descriptor Elements
   - COLOR SELECTION MODE (1 = direct)
   - EDGE WIDTH SPECIFICATION MODE (0 = absolute)
   - LINE WIDTH SPECIFICATION MODE (0 = absolute)
   - VDC EXTENT

d. Metafile Graphical Primitives with Associated Attributes

   Text Primitive Element with Attributes
   - TEXT COLOR
   - CHARACTER HEIGHT
   - TEXT FONT INDEX
   - CHARACTER ORIENTATION
   - TEXT

   Filled-Area Primitive Elements with Attributes
   - FILL COLOR
   - INTERIOR STYLE (1= solid or 4 = empty)
   - EDGE VISIBILITY (1 = on)
   - EDGE WIDTH
   - EDGE TYPE (1=solid or 2=dashed)
   - EDGE COLOR
   - POLYGON

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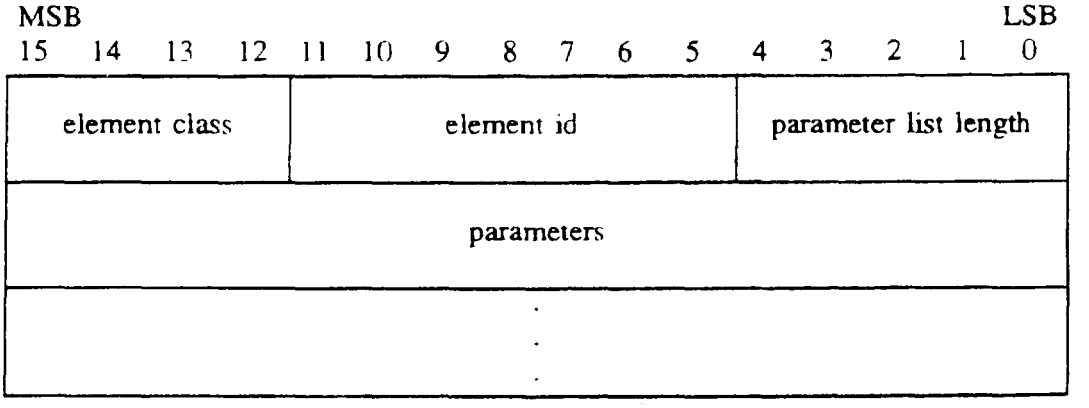
ELLIPSE
ELLIPTICAL ARC CLOSE
RECTANGLE
CIRCLE
CIRCULAR ARC CENTER CLOSE

Line Primitive Elements with Attributes

LINE WIDTH
LINE TYPE (1=solid or 2=dashed)
LINE COLOR
POLYLINE
ELLIPTICAL ARC
CIRCULAR ARC CENTER

4.2 CGM binary encoding. CGM commands in the CGM implementation for NITFS are encoded using the binary encoding method described in the FIPS PUB 128. Metafile elements will be represented in the binary encoding in either short-form commands or long-form commands. For short-form and long-form commands, the 4 most significant bits (MSB) identify the element class in which the command belongs (for example, Delimiter Elements, Descriptor Elements) and the fifth through eleventh bits identify the element ID (for example, BEGIN METAFILE, END METAFILE). For the short-form command, the five least significant bits (LSB) specify the length, in bytes, of the parameter list. This form is used to specify parameter lists that are less than 31 LONG. For the long-form command, the 5 LSB are set to the binary value "11111" (decimal 31). In this case, the next 2 bytes are interpreted as a signed positive integer containing the length of the parameter list. For short-form and long-form commands, the first byte of a string parameter specifies the length of the string. If necessary, parameters are padded with a trailing null byte (0) to ensure that all subsequent commands begin on a 16-bit word boundary. Note, the trailing null byte is not included in the parameter list length. In the figures contained in this standard, all numbers are decimal unless preceded by "0x" indicating hexadecimal notation.

TABLE 1. Short form of CGM command.



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TABLE 2. Long form of CGM command.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
element class				element id							31				
parameter list length															
parameters															
. . .															

4.3 CGM element flow. The following sequence of commands is used to describe all CGM graphics required for the CGM implementation for NITFS . The following flow diagram displays all the CGM commands for the NITFS implementation that could be contained in a metafile. The commands enclosed in brackets are optional within a given metafile and are used only when needed to describe the desired graphical symbol. Commands following the BEGIN PICTURE BODY command may be repeated to describe multiple graphics within the same metafile. The commands are executed in sequential order.

BEGIN METAFILE

METAFILE VERSION
METAFILE ELEMENT LIST
METAFILE DESCRIPTION
[FONT LIST]

BEGIN PICTURE

COLOR SELECTION MODE
[EDGE WIDTH SPECIFICATION MODE]
[LINE WIDTH SPECIFICATION MODE]
VDC EXTENT

BEGIN PICTURE BODY

[TEXT COLOR]
[CHARACTER HEIGHT]
[TEXT FONT INDEX]
[CHARACTER ORIENTATION]
[TEXT]

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[FILL COLOR]
[INTERIOR STYLE]
[EDGE VISIBILITY]
[EDGE WIDTH]
[EDGE TYPE]
[EDGE COLOR]
[POLYGON]
[ELLIPSE]
[ELLIPTICAL ARC CLOSE]
[RECTANGLE]
[CIRCLE]
[CIRCULAR ARC CENTER CLOSE]

[LINE WIDTH]
[LINE TYPE]
[LINE COLOR]
[POLYLINE]
[ELLIPTICAL ARC]
[CIRCULAR ARC CENTER]

END PICTURE

END METAFILE

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5. DETAILED REQUIREMENTS

5.1 Interface requirements. The following subsections list the required CGM commands along with the binary encoding method as described in the FIPS PUB 128 document. The words "command" and "element" are used synonymously throughout this document.

5.1.1 CGM interface input requirements.

5.1.1.1 Delimiter elements. The Delimiter elements define boundaries for significant structures within the metafile.

5.1.1.1.1 Begin Metafile element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Begin Metafile element using the following formats. The Begin Metafile element name is represented using the character string C1, C2, ... Cn with length n.

TABLE 3. Begin Metafile padded, short form input.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 0 LSB
0	1 parameter list length
length = n (even)	C1
C2	. . .
Cn	0

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Check the source to verify that this is the current version before use.

TABLE 4. Begin Metafile nonpadded, short form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
0				1							parameter list length				
length = n (odd)								C1							
C2								. . .							
C(n-1)								Cn							

TABLE 5. Begin Metafile padded, long form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				1							31					= 0x003F
length = n (even)								C1								
C2								. . .								
Cn								0								

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Check the source to verify that this is the current version before use.

TABLE 6. Begin Metafile nonpadded, long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
0	1	31	= 0x003F
parameter list length			
length = n (odd)	C1		
C2	. . .		
C(n-1)	Cn		

5.1.1.1.2 Begin Picture element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Begin Picture element using the following formats. The Begin Picture element name is represented using the character string C1, C2, ... Cn with length n.

TABLE 7. Begin Picture padded, short form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
0	3	parameter list length
length = n (even)	C1	
C2	. . .	
Cn	0	

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TABLE 8. Begin Picture nonpadded, short form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
0				3							parameter list length				
length = n (odd)								C1							
C2								. . .							
C(n-1)								0							

TABLE 9. Begin Picture padded, long form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				3							31					= 0x007F
length = n (even)								C1								
C2								. . .								
Cn								0								

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:58Z
Check the source to verify that this is the current version before use.

TABLE 10. Begin Picture nonpadded, long form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				3							31					= 0x007F
parameter list length																
length = n (odd)								C1								
C2								.								
								.								
C(n-1)								Cn								

5.1.1.1.3 Begin Picture Body element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Begin Picture Body element using the following format.

TABLE 11. Begin Picture Body input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				4							0					= 0x0080

5.1.1.1.4 End Picture element input. The CGM implementation for NITFS shall provide the capability to input and interpret the End Picture element using the following format.

TABLE 12. End Picture input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				5							0					= 0x00A0

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Check the source to verify that this is the current version before use.

5.1.1.1.5 End Metafile element input. The CGM implementation for NITFS shall provide the capability to input and interpret the End Metafile element using the following format.

TABLE 13. End Metafile input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
0				2							0					= 0x0040

5.1.1.2 Metafile Descriptor elements. The Metafile Descriptor elements describe the functional content, default conditions, and characteristics of the Metafile.

5.1.1.2.1 Metafile Version element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Metafile Version element (version 1) using the following format.

TABLE 14. Metafile Version input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
1				1							2					= 0x1022
1																= 0x0001

5.1.1.2.2 Metafile Description element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Metafile Description element using the following formats. The Metafile Description element name is represented using the character string C1, C2, ... Cn with length n.

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Check the source to verify that this is the current version before use.

TABLE 15. Metafile Description padded, short form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
1				2							parameter list length				
length = n (even)								C1							
C2								. . .							
Cn								0							

TABLE 16. Metafile Description nonpadded, short form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
1				2							parameter list length				
length = n (odd)								C1							
C2								. . .							
C(n-1)								Cn							

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Check the source to verify that this is the current version before use.

TABLE 17. Metafile Description padded, long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	2	31	= 0x105F
parameter list length			
length = n (even)	C1		
C2	. . .		
Cn	0		

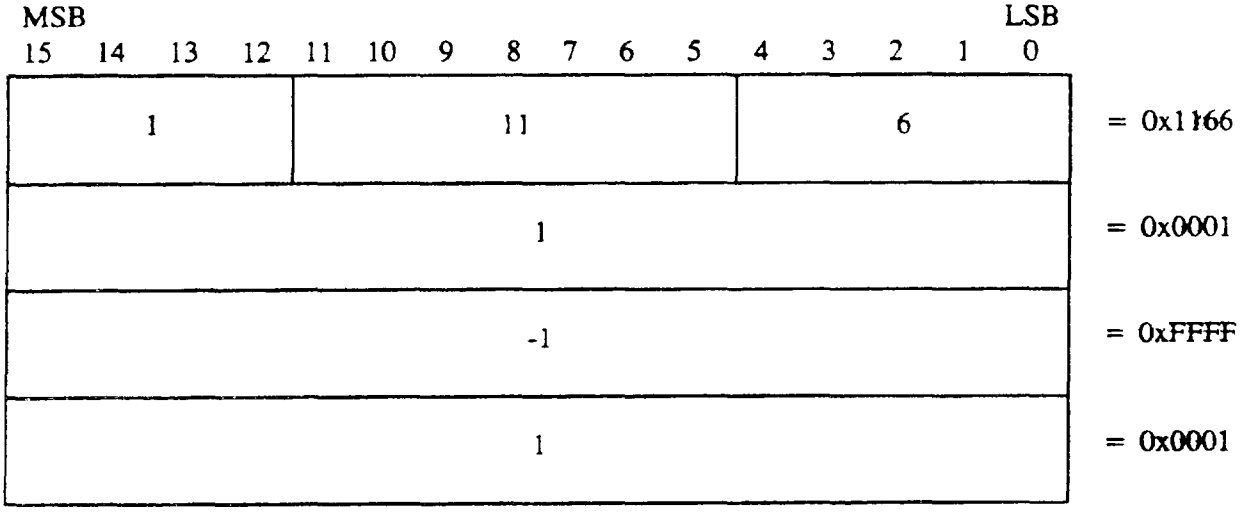
TABLE 18. Metafile Description nonpadded, long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	2	31	= 0x105F
parameter list length			
length = n (odd)	C1		
C2	. . .		
C(n-1)	Cn		

5.1.1.2.3 Metafile Element List element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Metafile Element List command using the following format. The metafile elements are chosen from specified CGM command subsets. When the second parameter is one, the metafile element is "Drawing Plus Control Set."

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Check the source to verify that this is the current version before use.

TABLE 19. Metafile Element List input.



5.1.1.2.4 Font List element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Font List element using the following formats. The first font name defined in the list is of length x with the font name given as the character string C1, C2, ... Cx assigned to index 1. The last font name defined in the list is of length z with the font name given as the character string C1, C2, ... Cz and assigned to index N. Selection of named fonts is accomplished with the Text Font Index element.

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TABLE 20. Font List padded, short form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
1	13	parameter list length
length = x	C1	
C2	. . .	
Cx	. . .	
length = z	C1	
C2	. . .	
Cz	0	

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TABLE 21. Font List nonpadded, short form input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
1				13							parameter list length				
length = x											C1				
C2											. . .				
C(x-1)											Cx				
. . .											length = z				
C1											C2				
. . .											Cz				

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TABLE 22. Font List padded, long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	13	31	= 0x11BF
parameter list length			
length = x	C1		
C2	. . .		
Cx	. . .		
length = z	C1		
C2	. . .		
Cz	0		

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TABLE 23. Font List nonpadded, long form input.

MSB 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 LSB 0		
1	13	31 = 0x11BF
parameter list length		
length = n	C1	
C2	. . .	
C(n-1)	Cn	
. . .	length = z	
C1	C2	
. . .	Cz	

5.1.1.3 Picture Descriptor elements. The Picture Descriptor elements set the interpretation modes of the attribute elements.

5.1.1.3.1 Color Selection Mode element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Color Selection Mode element using the following format.

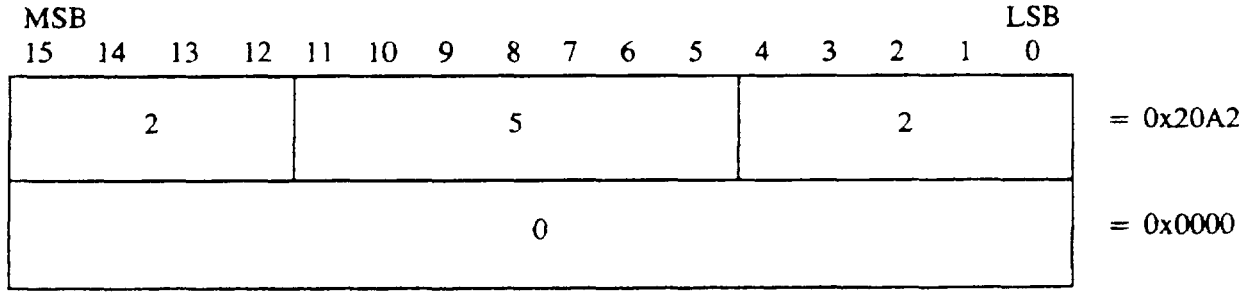
TABLE 24. Color Selection Mode input.

MSB 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 LSB 0			
2	2	2	= 0x2042
1			= 0x0001

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Check the source to verify that this is the current version before use.

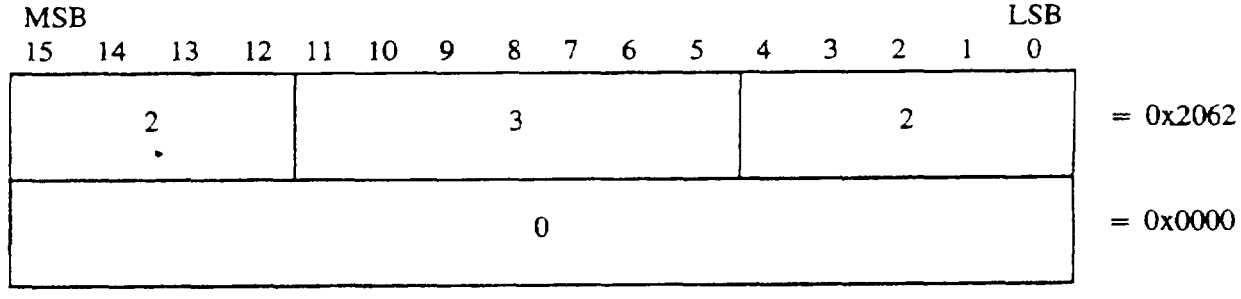
5.1.1.3.2 Edge Width Specification Mode element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Edge Width Specification Mode element using the following format. Edge width is specified in "absolute mode," which indicates that the width is specified in source pixels. This command is used to support filled-area primitives.

TABLE 25. Edge Width Specification Mode input.



5.1.1.3.3 Line Width Specification Mode element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Line Width Specification Mode element using the following format. Line width is specified in "absolute mode," which indicates that the width is specified in pixels. This command is used to support the line primitive.

TABLE 26. Line Width Specification Mode input.



5.1.1.3.4 VDC Extent element input. The CGM implementation for NITFS shall provide the capability to input and interpret the VDC Extent element using the following format.

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TABLE 27. VDC Extent input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
2				6							8					= 0x20C8
X1																
Y1																
X2																
Y2																

This element (illustrated on figures 2 and 3) defines the rectangular extent and orientation of the VDC space (the direction of the positive X and positive Y axes). The extent and orientation of VDC space is indicated by giving the coordinates of lower left hand and upper right hand corners of the VDC extent. The VDC extent space is a one-to-one map from the VDC extent to the source coordinates. The VDC extent origin (0,0) is mapped onto the source at the row and column location given by the SLOC field in the Symbol Subheader for a given image. The Symbol Subheader and the SLOC field are defined in the MIL-STD-2500 document. Note that X1 and X2 cannot be equal, and Y1 and Y2 cannot be equal. Note that the proper Character Orientation element must be present to ensure that CGM text is right side up for a given VDC Extent element. The VDC extent defines the positive 90-degree angle to be the right angle from the positive X-axis to the positive Y-axis.

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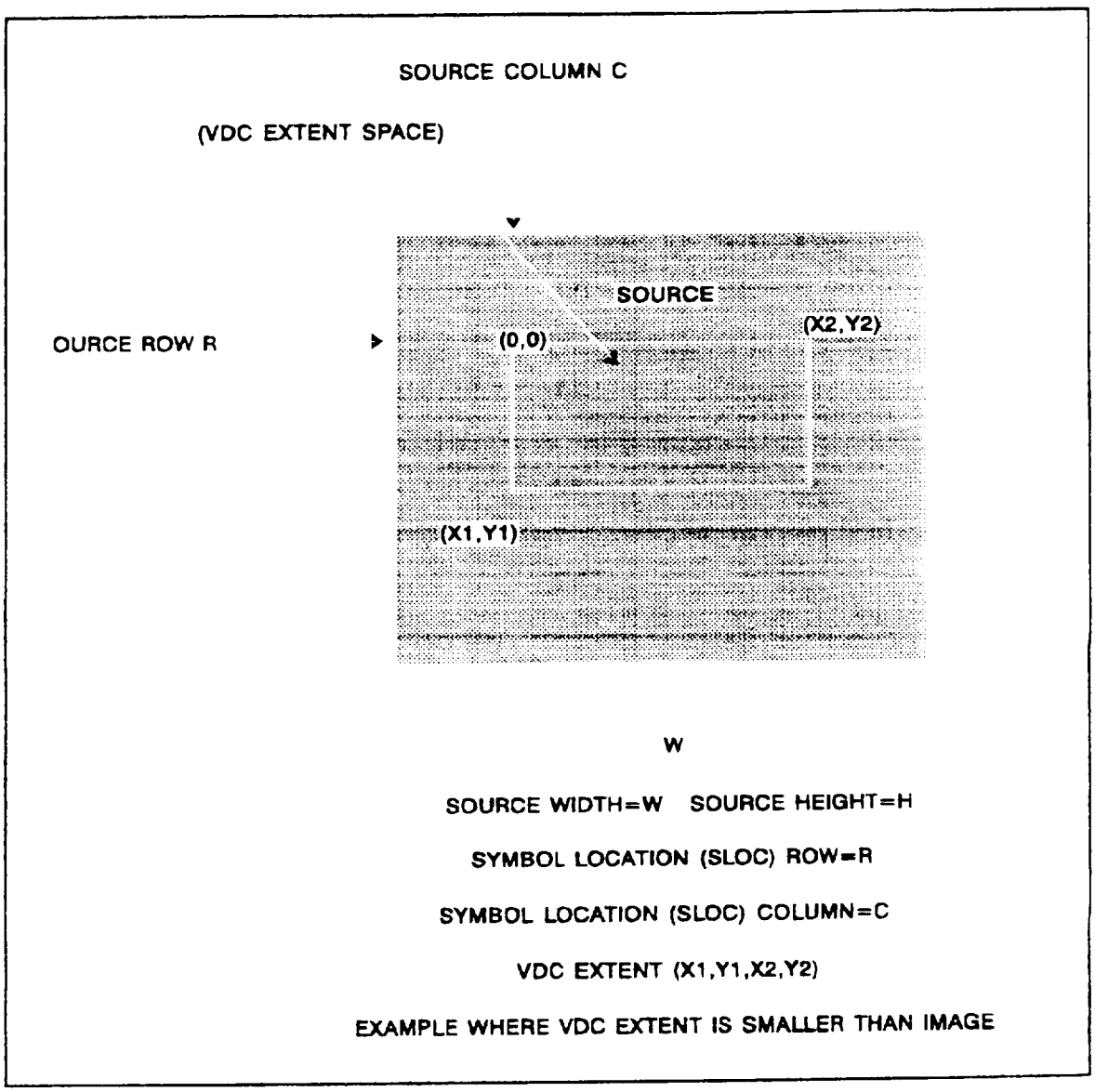


FIGURE 2. Small VDC Extent input.

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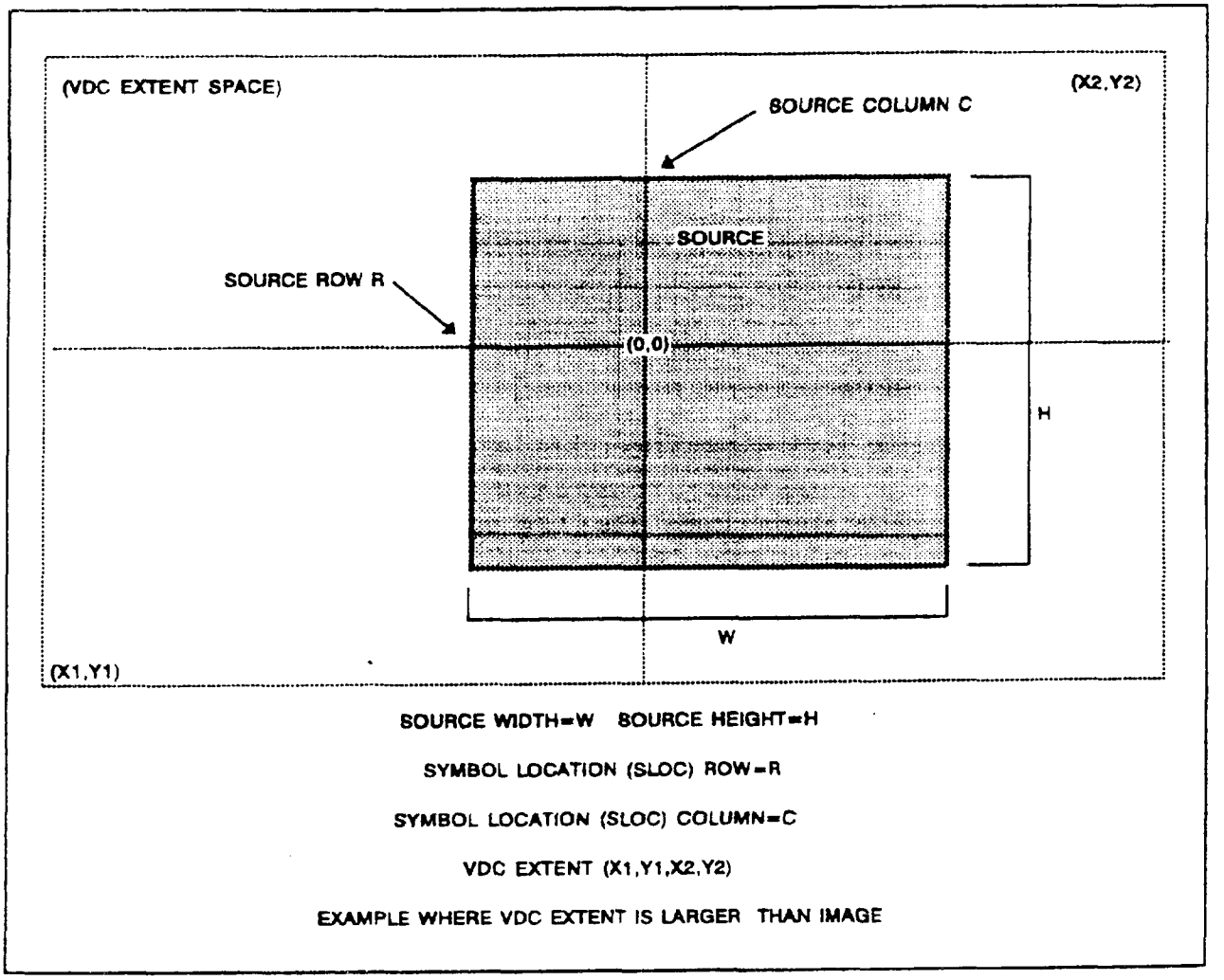


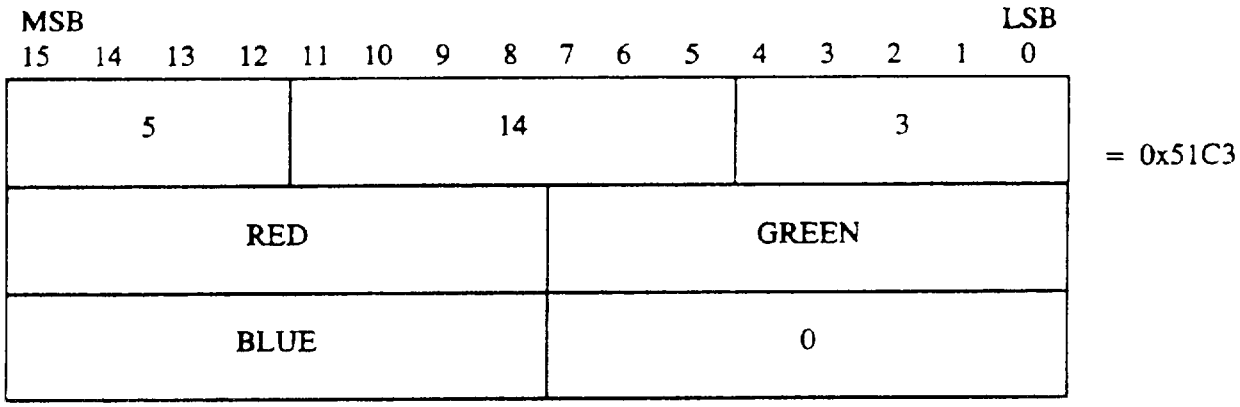
FIGURE 3. Large VDC Extent input.

5.1.1.4 Attribute elements. Attribute elements are used to describe the appearance of the Graphical Primitive elements.

5.1.1.4.1 Text Color element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Text Color element using the following format. The Text Color element is used to support the text primitives. Red, Green, and Blue (RGB) values are specified using a single byte, and the last byte of the command shall be a null byte.

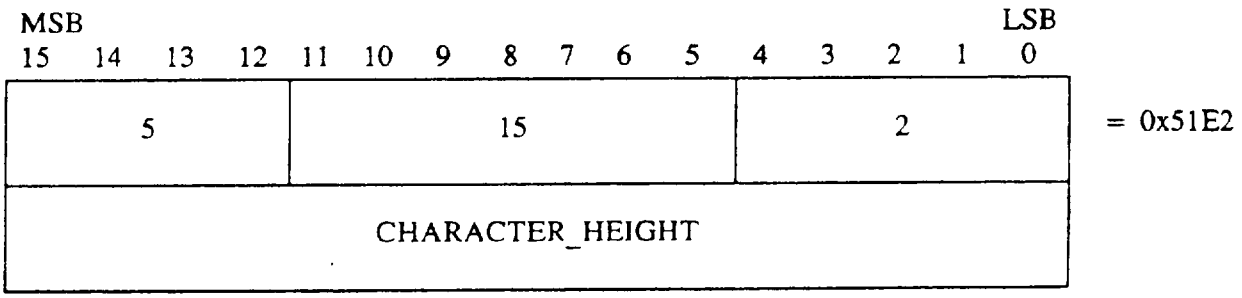
Source: https://assist.dla.mil -- Downloaded: 202640-26T07:58Z
Check the source to verify that this is the current version before use.

TABLE 28. Text Color input.



5.1.1.4.2 Character Height element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Character Height element using the following format. Variable character heights, as measured from baseline to capline, shall be supported. The font, if given, is specified by the Font List and Text Font Index elements.

TABLE 29. Character Height input.



5.1.1.4.3 Text Font Index element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Text Font Index element using the following format. The Text_Font_Index parameter is the index into the Font List array created from the Font List element for selection of a particular font.

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Check the source to verify that this is the current version before use.

TABLE 30. Text Font Index input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				10							2					= 0x5142
TEXT_FONT_INDEX																

5.1.1.4.4 Character Orientation element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Character Orientation element using the following format. This command defines the orientation and skew of the character body in subsequent text elements. Since the VDC Extent element is used to establish the direction of the positive X and positive Y axis, the character orientation must be present to ensure that text characters are always upright from left to right when the VDC Extent element defines Y increasing down or X increasing left. The following four cases apply to the Character Orientation element for each possible VDC extent orientation:

a. VDC Extent element with X increasing right and Y increasing up (X1 < X2 and Y1 < Y2).

   Character Orientation element not required or Y=1 and X=1 when Character Orientation present.

b. VDC Extent element with X increasing right and Y increasing down (X1 < X2 and Y1 > Y2).

   Character Orientation element required with Y=-1 and X=1.

c. VDC Extent element with X increasing left and Y increasing up (X1 > X2 and Y1 < Y2).

   Character Orientation element required with Y=1 and X=-1.

d. VDC Extent element with X increasing left and Y increasing down (X1 > X2 and Y1 > Y2).

   Character Orientation element required with Y=-1 and X=-1.

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TABLE 31. Character Orientation input.

MSB 15 14 13 12	11 10 9 8 7 6 5 4	3 2 1 0 LSB	
5	16	8	= 0x5208
0			= 0x0000
Y			
X			
0			= 0X0000

5.1.1.4.5 Fill Color element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Fill Color element using the following format. The Fill Color command is used to support the filled-area primitives. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

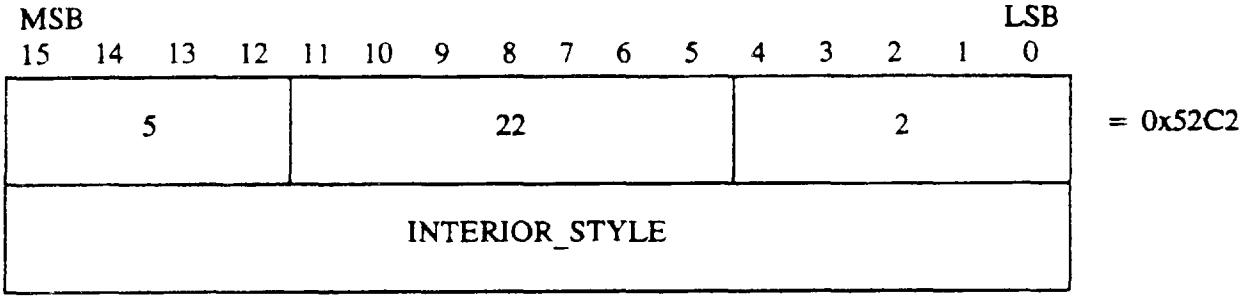
TABLE 32. Fill Color input.

MSB 15 14 13 12	11 10 9 8 7 6 5 4	3 2 1 0 LSB	
5	23	3	= 0x52E3
RED		GREEN	
BLUE		0	

5.1.1.4.6 Interior Style element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Interior Style element using the following format. Filled-area primitives must support the Interior_Style parameter for solid (1) and empty (4).

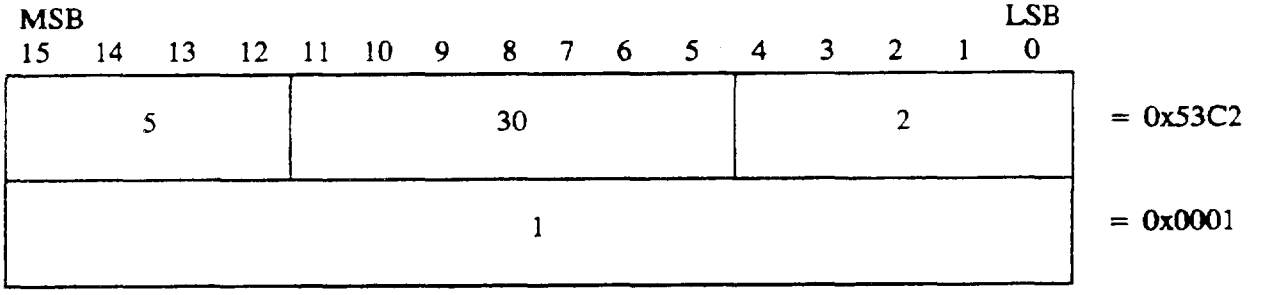
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

TABLE 33. Interior Style input.



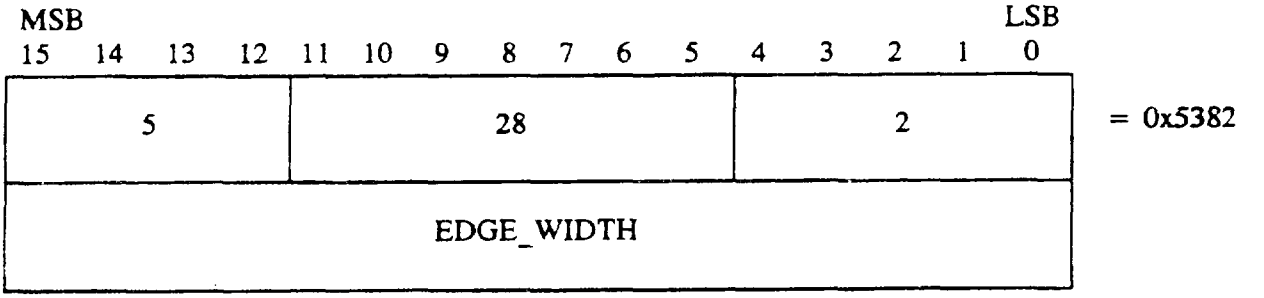
5.1.1.4.7 Edge Visibility element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Edge Visibility element using the following format. The edge visibility command (1 = on) is supported for use in filled-area primitives.

TABLE 34. Edge Visibility input.



5.1.1.4.8 Edge Width element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Edge Width element using the following format. Variable edge widths are supported for use in filled-area primitives.

TABLE 35. Edge Width input.



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5.1.1.4.9 Edge Type element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Edge Type element using the following format. The Edge_Type parameter can be solid (1) or dashed (2).

TABLE 36. Edge Type input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				27							2					= 0x5362
EDGE_TYPE																

5.1.1.4.10 Edge Color element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Edge Color element using the following format. Edge Color is supported for use in filled-area primitives. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

TABLE 37. Edge Color input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				29							3					= 0x53A3
RED								GREEN								
BLUE								0								

5.1.1.4.11 Line Width element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Line Width element using the following format. Variable line widths are supported for use by line primitives.

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TABLE 38. Line Width input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				3							2					= 0x5062
LINE_WIDTH																

5.1.1.4.12 Line Type element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Line Type element using the following format. The Line_Type parameter can be solid (1) or dashed (2).

TABLE 39. Line Type input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				2							2					= 0x5042
LINE_TYPE																

5.1.1.4.13 Line Color element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Line Color element using the following format. Line colors are supported for use by the line primitive. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

TABLE 40. Line Color input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				4							3					= 0x5083
RED								GREEN								
BLUE								0								

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5.1.1.5 Graphical Primitive elements. The Graphical Primitive elements describe the visual components of a picture as contained in the CGM.

5.1.1.5.1 Text element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Text element using the following formats. The X and Y parameters give the coordinates of the lower left hand corner of the text string. The flag parameter is an integer (1) indicating that this is the final text in the string. The length parameter is an unsigned byte containing the number of characters in the string. Finally, the text is given as a character string C1, C2, ... Cn with length n. If necessary, the last byte is padded with a zero so that the next command begins on a word boundary.

TABLE 41. Text padded, short form input.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	parameter list length		
X				
Y				
1				= 0x0001
length = n	C1			
C2	. . .			
Cn	0			

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TABLE 42. Text nonpadded, short form input.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	parameter list length		
X				
Y				
1				= 0x0001
length = n (odd)		C1		
C2		. . .		
C(n-1)		Cn		

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TABLE 43. Text padded, long form input.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	31		= 0x409F
parameter list length				
X				
Y				
1				= 0x0001
length = n (even)		C1		
C2		. . .		
Cn		0		

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TABLE 44. Text nonpadded, long form input.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4		31	= 0x409F
parameter list length				
X				
Y				
1				= 0x0001
length = n (odd)		C1		
C2		.		
C(n-1)		Cn		

5.1.1.5.2 Polygon element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Polygon element using the following format. The polygon parameters consist of a list of pairs of coordinates indicating the vertices of a polygon. The first vertex (Vertex1X, Vertex1Y) is connected to the last (Vertex(N)X, Vertex(N)Y) to close the polygon. Polygons are not "clipped" to the image boundary; therefore, some coordinates may specify off-image or off-screen locations, including negative locations. Note, the parameter list length is given as the total number of bytes for all vertex parameters (4N).

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TABLE 45. Polygon short form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
4	7	parameter list length (4N)
Vertex1X		
Vertex1Y		
Vertex2X		
Vertex2Y		
. . .		
Vertex(N)X		
Vertex(N)Y		

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TABLE 46. Polygon long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	7	31	= 0x40FF
parameter list length (4N)			
Vertex1X			
Vertex1Y			
Vertex2X			
Vertex2Y			
.			
Vertex(N)X			
Vertex(N)Y			

5.1.1.5.3 Ellipse element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Ellipse element using the following format. The ellipse parameters consist of a list of three pairs of coordinates. The first pair, CenterX and CenterY, give the location of the center of the ellipse. The other two pairs, End1X, End1Y, and End2X, End2Y, specify the end points of the first and second conjugate diameters. The ellipse interior will be as specified by the Interior Style command, and the edge width, type, and color will be as specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

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TABLE 47. Ellipse input.

MSB LSB

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
4				17							12					= 0x422C
CenterX																
CenterY																
End1X																
End1Y																
End2X																
End2Y																

5.1.1.5.4 Polyline element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Polyline element using the following format. The polyline parameters consist of a list of pairs of coordinates indicating the vertices of a polyline (Vertex1X, Vertex1Y) to (Vertex(N)X, Vertex(N)Y). The line width, type, and color are specified by the Line Width, Line Type, and Line Color commands, respectively. Polylines are not "clipped" to the image boundary. Therefore, some coordinates may specify off-image or off-screen locations, including negative locations. Note, the parameter list length is given as the total number of bytes for all vertex parameters (4N).

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TABLE 48. Polyline short form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
4	1	parameter list length (4N)
Vertex1X		
Vertex1Y		
Vertex2X		
Vertex2Y		
. . .		
Vertex(N)X		
Vertex(N)Y		

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TABLE 49. Polyline long form input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	1	31	= 0x403F
parameter list length (4N)			
Vertex1X			
Vertex1Y			
Vertex2X			
Vertex2Y			
.			
.			
.			
Vertex(N)X			
Vertex(N)Y			

5.1.1.5.5 Elliptical Arc element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Elliptical Arc element using the following format. A conjugate diameter pair of an ellipse is a pair of diameters of the ellipse such that a tangent to the ellipse at each endpoint is parallel to the other diameter. The centerpoint (CenterX, CenterY) specifies the center of an ellipse. The conjugate diameter endpoints (End1X, End1Y, and End2X, End2Y) include one endpoint from each conjugate diameter; together with the centerpoint they define the two conjugate diameters of the ellipse. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. A start ray and end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors respectively. The define arc begins at the intersection of the ellipse and the end ray in the direction defined as follows. A "conjugate radius" is defined to be half of a conjugate diameter. Letting the centerpoint be labeled M, the first conjugate diameter endpoint P_1, and the second conjugate diameter endpoint P_2, then the line segments M-P_1 and M-P_2 define two conjugate radii, referred to in what follows as the first conjugate radius and the second conjugate

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radius respectively. The conjugate radii meet at M and define two angles: the sum of the two angles is 360 degrees, one angle is less than 180 degrees and the other is greater than 180 degrees. The drawing direction of the elliptical arc is the direction from the first conjugate radius to the second conjugate radius through the smaller of these two angles. Valid values of the three specifying points of the ellipse are those which yield three distinct points. The specified ellipse is non-degenerate if and only if the three points are non-colinear. Valid values of the vector components are those which produce vectors of non-zero length. If the start ray and end ray are coincident, it is ambiguous whether the defined arc is null (zero arc length) or the entire ellipse. The arc width, type and color are specified by the Line Width, Line Type and Line Color commands, respectively.

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TABLE 50. Elliptical Arc input.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 0 LSB	
4	18	20 = 0x4254
CenterX		
CenterY		
End1X		
End1Y		
End2X		
End2Y		
StartVectorX		
StartVectorY		
EndVectorX		
EndVectorY		

5.1.1.5.6 Elliptical Arc Close element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Elliptical Arc Close element using the following format. A conjugate diameter pair of an ellipse is a pair of diameters of the ellipse such that the tangent to the ellipse at each endpoint is parallel to the other diameter. The centerpoint (CenterX, CenterY) specifies the center of an ellipse. The conjugate diameter endpoints (End1X, End1Y, and End2X, End2Y) include one endpoint from each conjugate diameter; together with the centerpoint they define the two conjugate diameters of the ellipse. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. A start ray and end ray are derived from the start and end

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vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The define arc begins at the intersection of the ellipse and the end ray in the direction defined as follows. A "conjugate radius" is defined to be half of a conjugate diameter. Letting the centerpoint be labeled M, the first conjugate diameter endpoint P_1, and the second conjugate diameter endpoint P_2, then the line segments M-P_1 and M-P_2 define two conjugate radii, referred to in what follows as the first conjugate radius and the second conjugate radius, respectively. The conjugate radii meet at M and define two angles: the sum of the two angles is 360 degrees, one angle is less than 180 degrees and the other is greater that 180 degrees. The drawing direction of the elliptical arc is the direction from the first conjugate radius to the second conjugate radius through the smaller of these two angles. Valid values of the three specifying points of the ellipse are those which yield three distinct points. The specified ellipse is non-degenerate if and only if the three points are non-colinear. Valid values of the vector components are those which produce vectors of non-zero length. If the start ray and end ray are coincident, it is ambiguous whether the defined arc is null (zero arc length) or the entire ellipse. The last parameter, Close Type, specifies how the arc is closed (0=pie or 1=chord). If the close type is chord, a line is drawn between the endpoints of the arc. If the close type is pie, a line is drawn from the beginning of the arc to the centerpoint of the ellipse and then to the end of the arc. The interior of the arc will be specified by the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

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TABLE 51. Elliptical Arc Close input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	19	22	= 0x4276
CenterX			
CenterY			
End1X			
End1Y			
End2X			
End2Y			
StartVectorX			
StartVectorY			
EndVectorX			
EndVectorY			
Close Type			

5.1.1.5.7 Rectangle element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Rectangle element using the following format. The rectangle is defined by two distinct points that are diagonal opposite corners of the rectangle where the rectangle is oriented parallel to the VDC axes. The interior of the rectangle will be as specified by the Interior Style command, and the edge width, type and color will be as specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

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TABLE 52. Rectangle input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
4				11							8					= 0x4168
Corner1X																
Corner1Y																
Corner2X																
Corner2Y																

5.1.1.5.8 Circle element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Circle element using the following format. The first two parameters, CenterX and CenterY, give the location of the center of the circle. The next parameter, Radius, specifies the radius of the circle. Only non-negative values are valid for the radius. The interior of the circle will be specified by the Interior Style command, and the edge width, type and color will be as specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

TABLE 53. Circle input.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
4				12							6					= 0x4186
CenterX																
CenterY																
Radius																

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5.1.1.5.9 Circular Arc Center element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Circular Arc Center element using the following format. The specified radius (Radius) and centerpoint (CenterX, CenterY) define a circle. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. The start ray and end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors respectively. The arc is drawn from the intersection of the circle and the start ray to the intersection of the circle, and the end ray in the positive angular direction, as defined by the VDC Extent. Valid values of the vector components are those which produce distinct vectors on non-zero length. The arc width, type and color are specified by the Line Width, Line Type and Line Color commands, respectively.

TABLE 54. Circular Arc Center input.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	15	14	= 0x41EE
CenterX			
CenterY			
StartX			
StartY			
EndX			
EndY			
Radius			

5.1.1.5.10 Circular Arc Center Close element input. The CGM implementation for NITFS shall provide the capability to input and interpret the Circular Arc Center Close element using the following format. The specified radius (Radius) and centerpoint (CenterX, CenterY) define a circle. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. The start ray and

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end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The arc is drawn from the intersection of the circle and the start ray to the intersection of the circle and the end ray in the positive angular direction, as defined by the VDC Extent. The last parameter, CloseType, specifies how the arc is closed (0=pie or 1=chord). If the close type is chord, a line is drawn between the endpoints of the arc. If the close type is pie, a line is drawn from the starting point through the computed arc center to the ending point. Valid values of the vector components are those which produce distinct vectors of non-zero length. The interior of this element will be as specified by the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type and Edge Color commands, respectively.

TABLE 55. Circular Arc Center Close input.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
4	16	16	= 0x4210
CenterX			
CenterY			
StartX			
StartY			
EndX			
EndY			
Radius			
Close Type			

5.1.1.6 CGM binary encoding. The CGM elements are represented in the binary encoding format as described in the FIPS PUB 128 document.

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5.1.1.6.1 CGM binary encoding for input. The CGM implementation for NITFS shall provide the capability to input and interpret the CGM elements in the binary encoding format.

5.1.2 CGM interface output requirements.

5.1.2.1 Delimiter elements. The Delimiter elements define boundaries for significant structures within the metafile.

5.1.2.1.1 Begin Metafile element output. The CGM implementation for NITFS shall provide the capability to generate and output the Begin Metafile element with the following formats. The Begin Metafile element name is represented using the character string C1, C2, ... Cn with length n.

TABLE 56. Begin Metafile padded, short form output.

MSB LSB

15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0
0	1	parameter list length
length = n (even)		C1
C2		. . .
Cn		0

TABLE 57. Begin Metafile nonpadded, short form output.

MSB LSB

15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0
0	1	parameter list length
length = n (odd)		C1
C2		. . .
C(n-1)		Cn

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TABLE 58. Begin Metafile padded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
0	1	31	= 0x003F
parameter list length			
length = n (even)	C1		
C2	. . .		
Cn	0		

TABLE 59. Begin Metafile nonpadded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
0	1	31	= 0x003F
parameter list length			
length = n (odd)	C1		
C2	. . .		
C(n-1)	Cn		

5.1.2.1.2 Begin Picture element output. The CGM implementation for NITFS shall provide the capability to generate and output the Begin Picture element with the following formats. The Begin Picture element name is represented using the character string C1, C2, ... Cn with length n.

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TABLE 60. Begin Picture padded, short form output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
0								3							parameter list length
length = n (even)															C1
C2															. . .
Cn															0

TABLE 61. Begin Picture nonpadded, short form output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
0								3							parameter list length
length = n (odd)															C1
C2															. . .
C(n-1)															Cn

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Check the source to verify that this is the current version before use.

TABLE 62. Begin Picture padded, long form output.

MSB 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 LSB

0	3	31	= 0x007F
parameter list length			
length = n (even)	C1		
C2	. . .		
Cn	0		

TABLE 63. Begin Picture nonpadded, long form output.

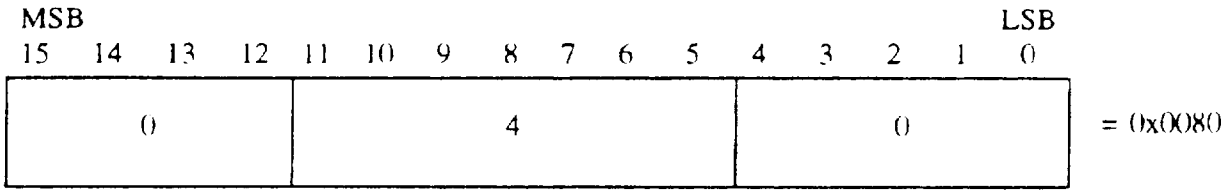
MSB 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 LSB

0	3	31	= 0x007F
parameter list length			
length = n (odd)	C1		
C2	. . .		
C(n-1)	Cn		

5.1.2.1.3 Begin Picture Body element output. The CGM implementation for NITFS shall provide the capability to generate and output the Begin Picture Body element using the following format.

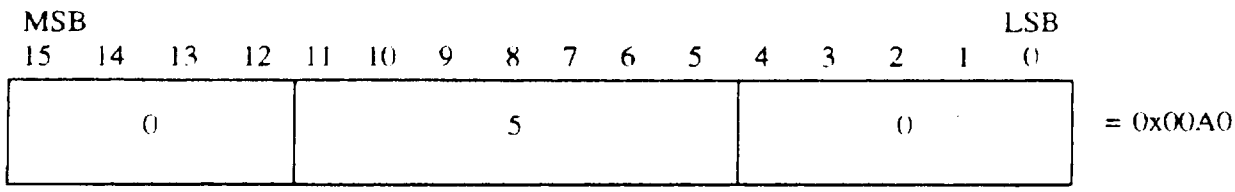
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:58Z
Check the source to verify that this is the current version before use.

TABLE 64. Begin Picture Body output.



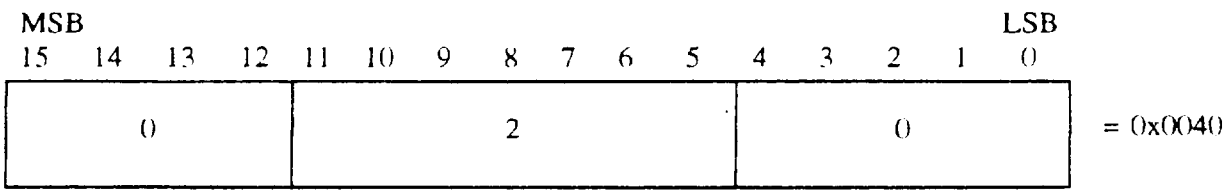
5.1.2.1.4 End Picture element output. The CGM implementation for NITFS shall provide the capability to generate and output the End Picture element using the following format.

TABLE 65. End Picture output.



5.1.2.1.5 End Metafile element output. The CGM implementation for NITFS shall provide the capability to generate and output the End Metafile element using the following format.

TABLE 66. End Metafile output.



5.1.2.2 Metafile Descriptor elements. The Metafile Descriptor elements describe the functional content, default conditions, and characteristics of the Metafile.

5.1.2.2.1 Metafile Version element output. The CGM implementation for NITFS shall provide the capability to generate and output the Metafile Version element (version 1) using the following format.

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Check the source to verify that this is the current version before use.

TABLE 67. Metafile Version output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	1	2	= 0x1022
1			= 0x0001

5.1.2.2.2 Metafile Description element output. The CGM implementation for NITFS shall provide the capability to generate and output the Metafile Description element using the following formats. The Metafile Description element name is represented using the character string C1, C2, ... Cn with length n.

TABLE 68. Metafile Description padded, short form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
1	2	parameter list length
length = n (even)	C1	
C2	. . .	
Cn	0	

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TABLE 69. Metafile Description nonpadded, short form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0
1	2	parameter list length
length = n (odd)	C1	
C2	. . .	
C(n-1)	Cn	

TABLE 70. Metafile Description padded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
1	2	31	= 0x105F
parameter list length			
length = n (even)	C1		
C2	. . .		
Cn	0		

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TABLE 71. Metafile Description nonpadded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 LSB 0	
1	2	31 = 0x105F
parameter list length		
length = n (odd)	C1	
C2	. . .	
C(n-1)	Cn	

5.1.2.2.3 Metafile Element List element output. The CGM implementation for NITFS shall provide the capability to generate and output the Metafile Element List element using the following format. The subset of commands from which metafile elements are chosen is specified by the CGM defined as "Drawing Plus Control Set" when the second parameter is one.

TABLE 72. Metafile Element List output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 LSB 0	
1	11	6 = 0x1166
1		= 0x0001
-1		= 0xFFFF
1		= 0x0001

5.1.2.2.4 Font List element output. The CGM implementation for NITFS shall provide the capability to generate and output the Font List element using the following formats. The first font name defined in the list is of length x with the font name given as the character string C1, C2, ...

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Cx and assigned to index 1. The last font name defined in the list is of length z with the font name given as the character string C1, C2, ... Cz and assigned to index N. Named fonts are selected by using the Text Font Index element.

TABLE 73. Font List padded, short form output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 LSB 0
1	13
parameter list length	
length = x	C1
C2	. . .
Cx	. . .
length = z	C1
C2	. . .
Cz	0

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TABLE 74. Font List nonpadded, short form output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0
1											13				parameter list length
length = x											C1				
C2											. . .				
C(x-1)											Cx				
. . .											length = z				
C1											C2				
. . .											Cz				

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TABLE 75. Font List padded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	13	31	= 0x11BF
parameter list length			
length = x	C1		
C2	.		
Cx	.		
length = z	C1		
C2	.		
Cz	0		

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TABLE 76. Font List nonpadded, long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
1	13	31	= 0x11BF
parameter list length			
length = n		C1	
C2		. . .	
C(n-1)		Cn	
. . .		length = z	
C1		C2	
. . .		Cz	

5.1.2.3 Picture Descriptor elements. The Picture Descriptor Elements set the interpretation modes of the attribute elements.

5.1.2.3.1 Color Selection Mode element output. The CGM implementation for NITFS shall provide the capability to generate and output the Color Selection Mode element using the following format.

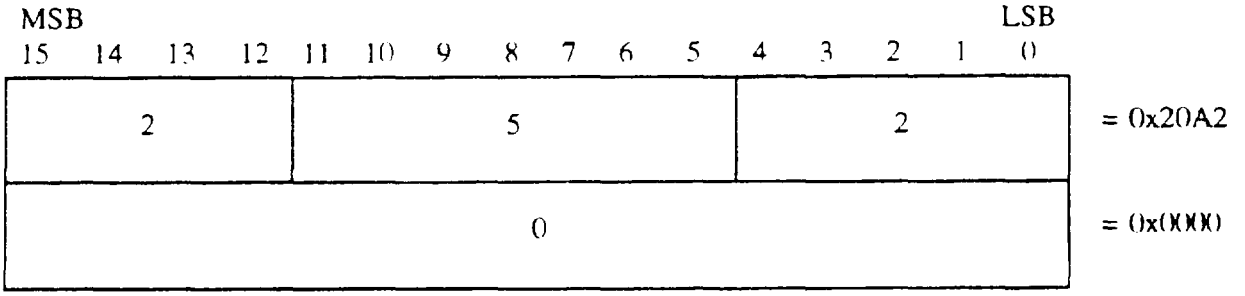
TABLE 77. Color Selection Mode output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
2	2	2	= 0x2042
1			= 0x0001

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Check the source to verify that this is the current version before use.

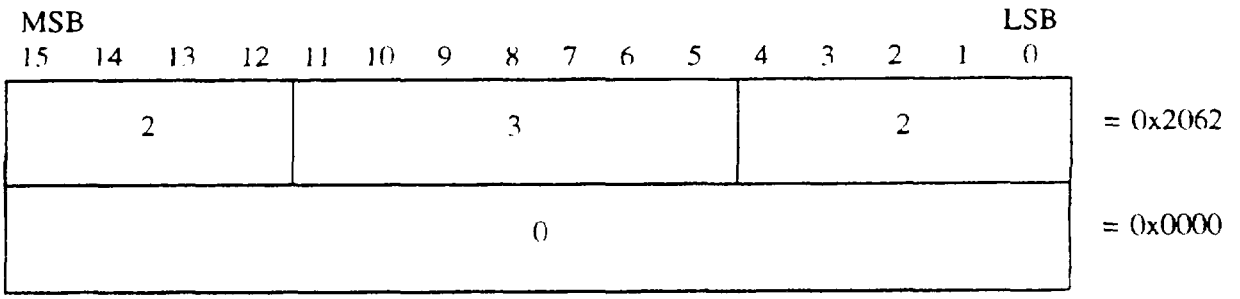
5.1.2.3.2 Edge Width Specification Mode element output. The CGM implementation for NITFS shall provide the capability to generate and output the Edge Width Specification Mode element using the following format. Edge width is specified in "absolute mode," which indicates that the width is specified in pixels. This command is used to support filled-area primitives.

TABLE 78. Edge Width Specification Mode output.



5.1.2.3.3 Line Width Specification Mode element output. The CGM implementation for NITFS shall provide the capability to generate and output the Line Width Specification Mode element using the following format. Line width is specified in "absolute mode," which indicates that the width is specified in pixels. This command is used to support the Polyline primitive.

TABLE 79. Line Width Specification Mode output.



5.1.2.3.4 VDC Extent element output. The CGM implementation for NITFS shall provide the capability to generate and output the VDC Extent element using the following format.

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TABLE 80. VDC Extent output.

MSB LSB

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
2				6							8					= 0x20C8
X1																
Y1																
X2																
Y2																

This element (illustrated on figures 4 and 5) defines the rectangular extent of the VDC space and the orientation of VDC space (the direction of the positive X and positive Y axes). The extent and orientation of VDC space is indicated by giving the coordinates of lower left hand and upper right hand corners of the VDC extent. The VDC extent space is a one-to-one map from the VDC extent to the source coordinates. The VDC extent origin (0,0) is mapped onto the source at the row and column location given by the SLOC field in the Symbol Subheader for a given image. The Symbol Subheader and the SLOC field are defined in the MIL-STD-2500 document. Note that X1 and X2 cannot be equal, and Y1 and Y2 cannot be equal. Note that the proper Character Orientation element must be present to ensure that CGM text is orientated right side up for a given VDC Extent element.

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Check the source to verify that this is the current version before use.

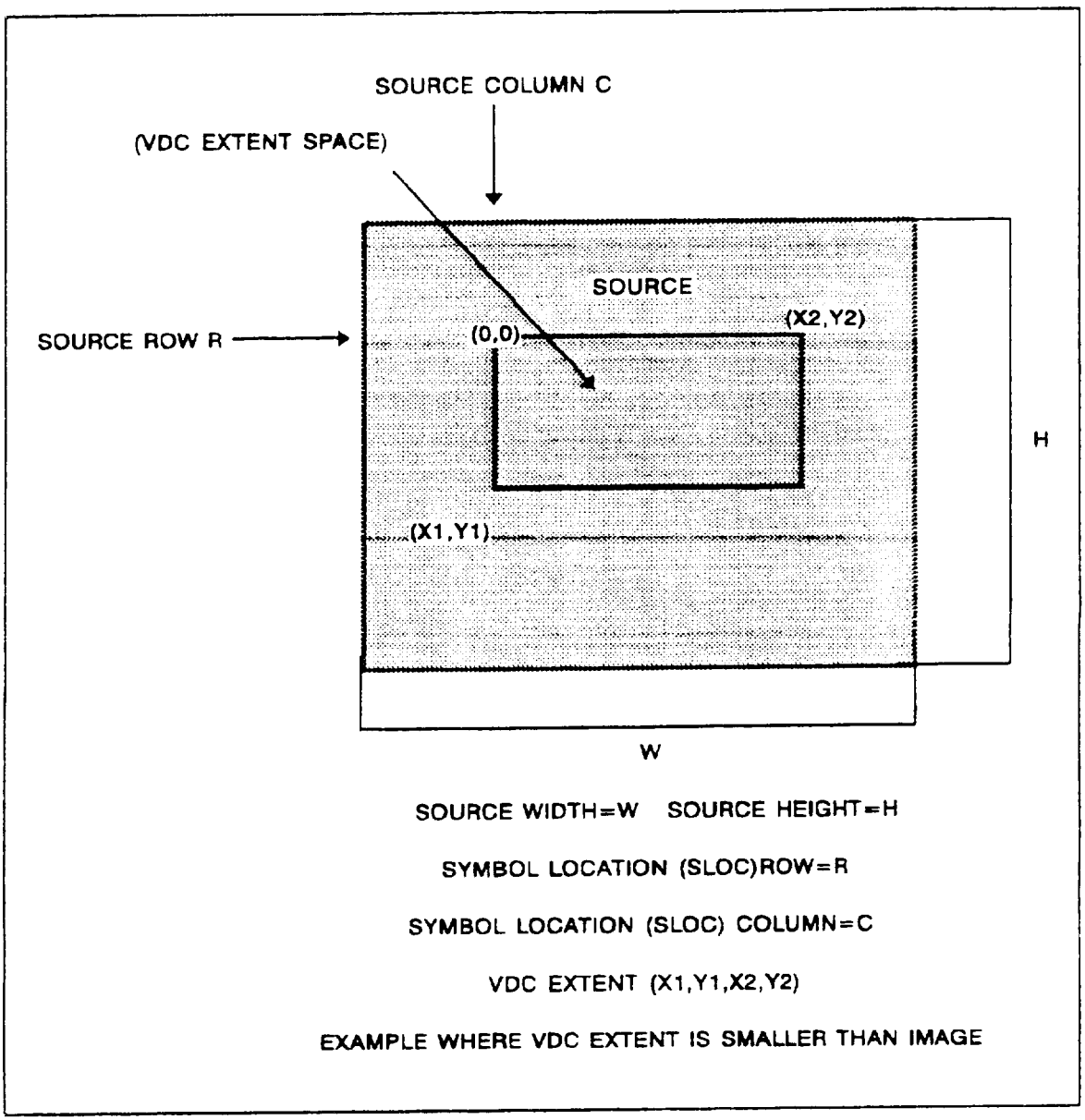


FIGURE 4. Small VDC Extent output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
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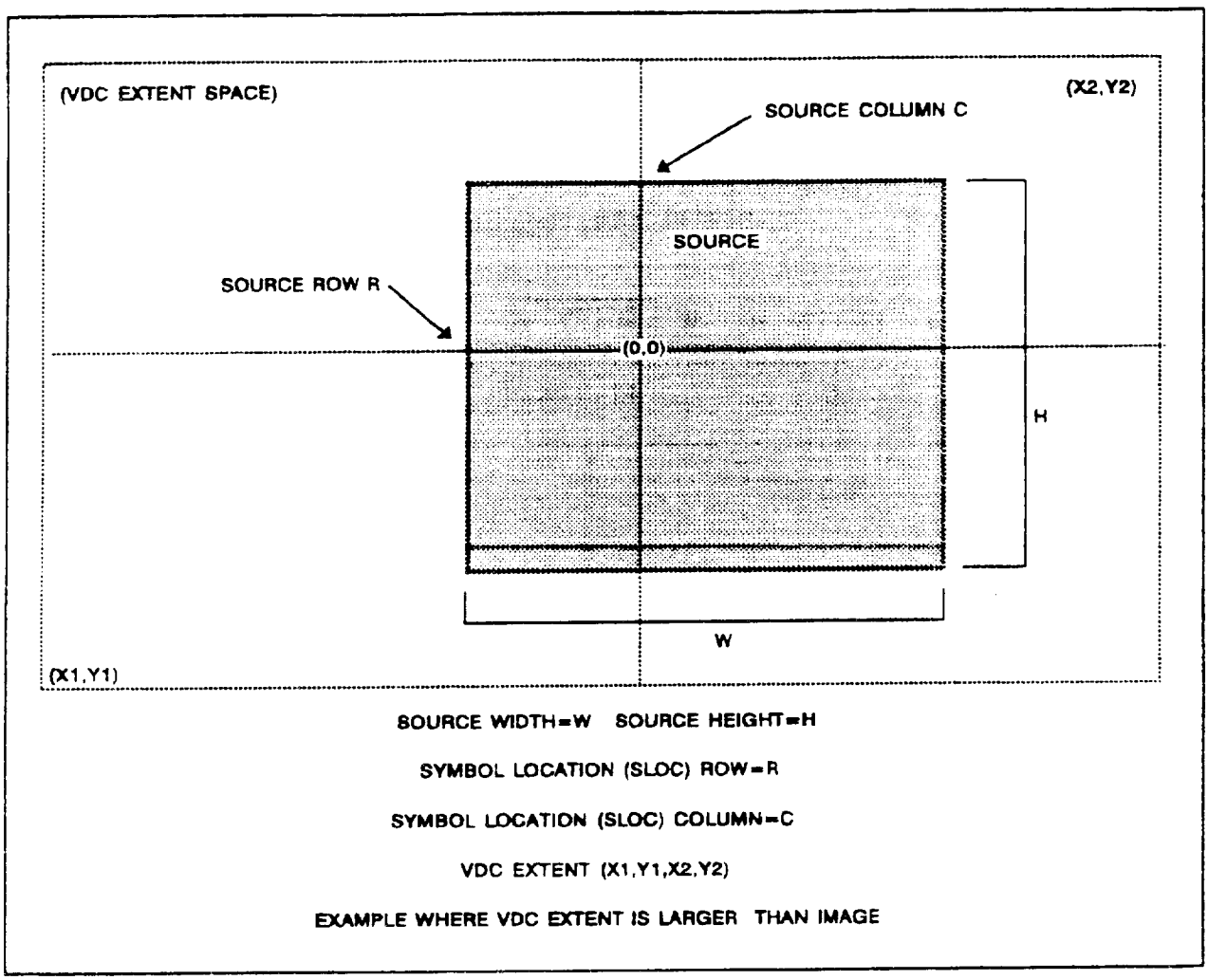


FIGURE 5. Large VDC Extent output.

5.1.2.4 Attribute elements. The Attribute elements are used to describe the appearance of the Graphical Primitive elements. Full color may be specified for the Elements dealing with color. Color items for receiving systems unable to support full color must be mapped to colors they are able to support.

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5.1.2.4.1 Text Color element output. The CGM implementation for NITFS shall provide the capability to generate and output the Text Color element using the following format. The Text Color element is used to support the text primitives. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

TABLE 81. Text Color output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				14							3					= 0x51C3
RED								GREEN								
BLUE								0								

5.1.2.4.2 Character Height element output. The CGM implementation for NITFS shall provide the capability to generate and output the Character Height element using the following format. As measured from baseline to capline, variable character heights shall be supported. If given, the font is specified by the Font List and Text Font Index elements.

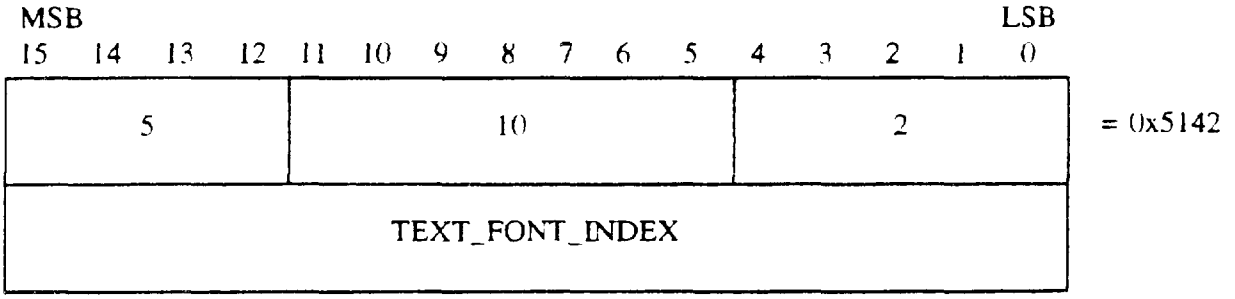
TABLE 82. Character Height output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				15							2					= 0x51E2
CHARACTER_HEIGHT																

5.1.2.4.3 Text Font Index element output. The CGM implementation for NITFS shall provide the capability to generate and output the Text Font Index element using the following format. The Text_Font_Index parameter is the index into the Font List array created from the Font List element for selection of a particular font.

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Check the source to verify that this is the current version before use.

TABLE 83. Text Font Index output.



5.1.2.4.4 Character Orientation element output. The CGM implementation for NITFS shall provide the capability to generate and output the Character Orientation element using the following format. This command defines the orientation and skew of the character body in subsequent text elements. Since the VDC Extent element is used to establish the direction of the positive X and positive Y axis, the character orientation must be present to ensure that text characters are always upright from left to right when the VDC Extent element defines Y increasing down or X increasing left. The following four cases apply to the Character Orientation element for each possible VDC extent orientation.

a. VDC Extent element with X increasing right and Y increasing up (X1<X2 and Y1<Y2).

   Character Orientation element not required or Y=1 and X=1 when Character Orientation present.

b. VDC Extent element with X increasing right and Y increasing down (X1<X2 and Y1>Y2).

   Character Orientation element required with Y=-1 and X=1.

c. VDC Extent element with X increasing left and Y increasing up (X1>X2 and Y1<Y2).

   Character Orientation element required with Y=1 and X=-1.

d. VDC Extent element with X increasing left and Y increasing down (X1>X2 and Y1>Y2).

   Character Orientation element required with Y=-1 and X=-1.

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TABLE 84. Character Orientation output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5							16						8			= 0x5208
0																= 0x0000
Y																
X																
0																= 0x0000

5.1.2.4.5 Fill Color element output. Provide the capability to generate and output the Fill Color element using the following format. The Fill Color command is used to support the filled-area primitives. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

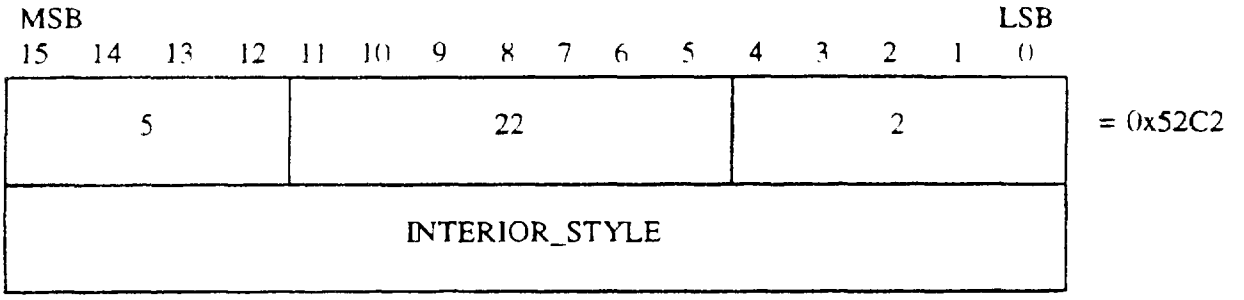
TABLE 85. Fill Color output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5							23						3			= 0x52E3
RED											GREEN					
BLUE											0					

5.1.2.4.6 Interior Style element output. The CGM implementation for NTTFS shall provide the capability to generate and output the Interior Style element using the following format. Filled-area primitives must support the Interior_Style parameter for solid (1) and empty (4).

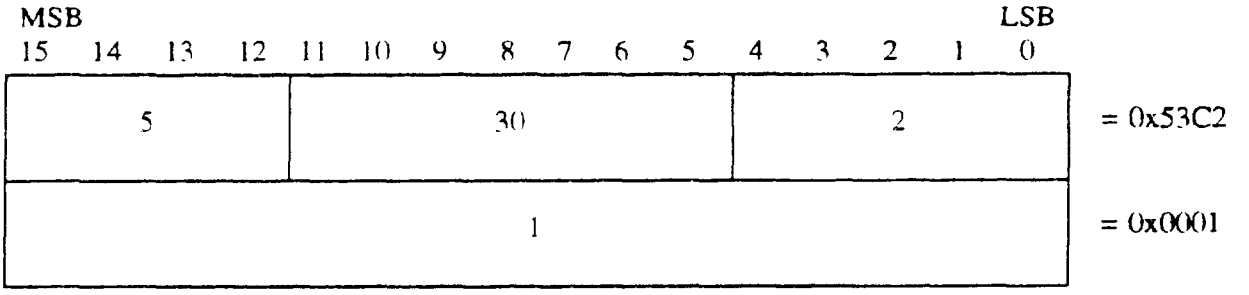
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

TABLE 86. Interior Style output.



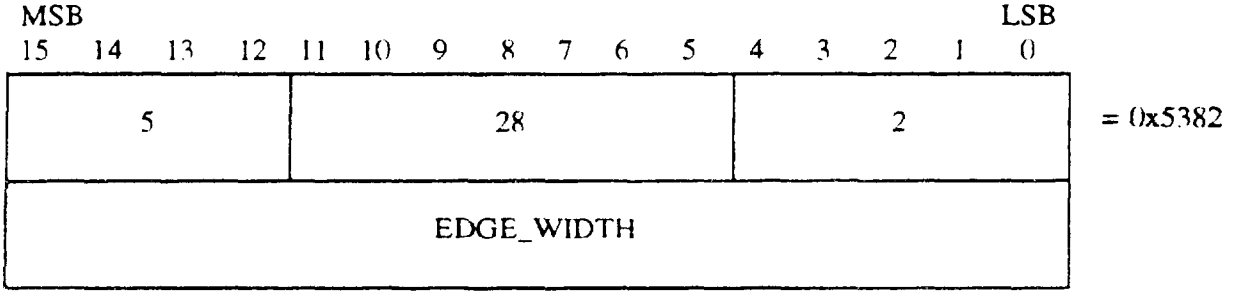
5.1.2.4.7 Edge Visibility element output. The CGM implementation for NITFS shall provide the capability to generate and output the Edge Visibility element using the following format. The edge visibility command (1 = on) is supported for use in filled-area primitives.

TABLE 87. Edge Visibility output.



5.1.2.4.8 Edge Width element output. The CGM implementation for NITFS shall provide the capability to generate and output the Edge Width element using the following format. Variable edge widths are supported for use in filled-area primitives.

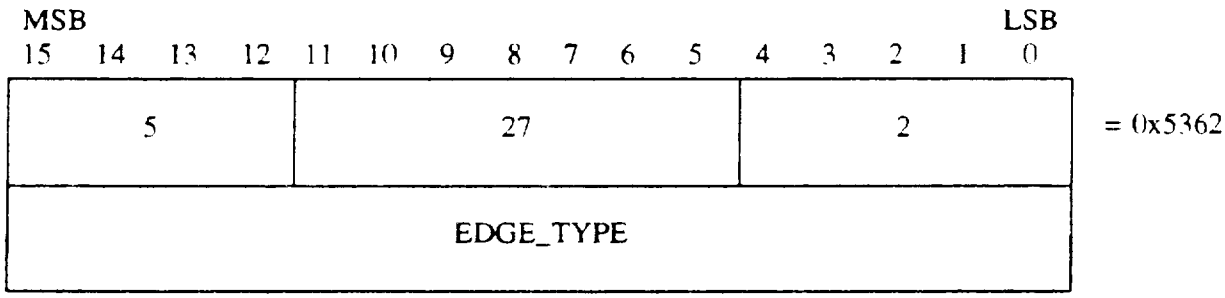
TABLE 88. Edge Width output.



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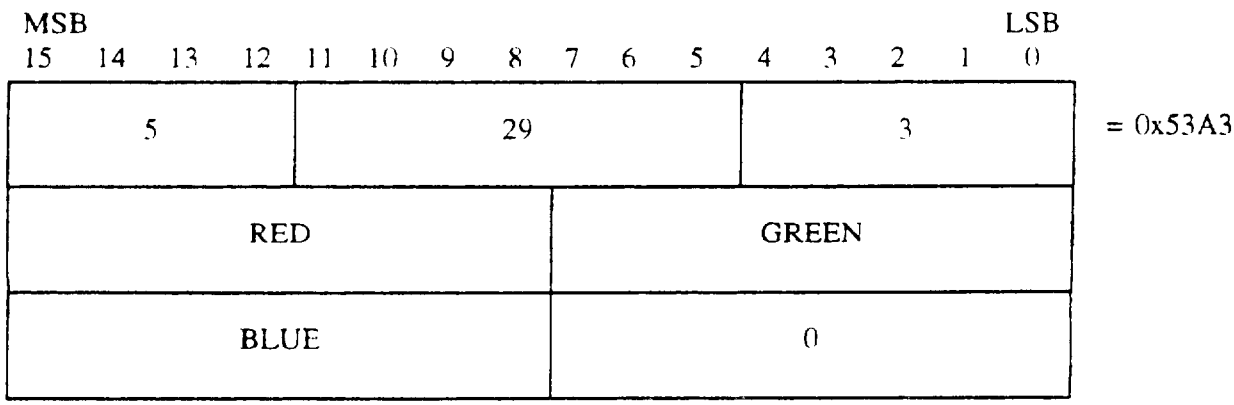
5.1.2.4.9 Edge Type element output. The CGM implementation for NITFS shall provide the capability to generate and output the Edge Type element using the following format. The Edge_Type parameter can be solid (1) or dashed (2).

TABLE 89. Edge Type output.



5.1.2.4.10 Edge Color element output. The CGM implementation for NITFS shall provide the capability to generate and output the Edge Color element using the following format. Edge color is supported for use in filled-area primitives. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

TABLE 90. Edge Color output.



5.1.2.4.11 Line Width element output. The CGM implementation for NITFS shall provide the capability to generate and output the Line Width element using the following format. Variable line widths are supported for use by line primitives.

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TABLE 91. Line Width output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				3							2					= 0x5062
LINE_WIDTH																

5.1.2.4.12 Line Type element output. The CGM implementation for NITFS shall provide the capability to generate and output the Line Type element using the following format. The Line_Type parameter can be solid (1) or dashed (2).

TABLE 92. Line Type output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				2							2					= 0x5042
LINE_TYPE																

5.1.2.4.13 Line Color element output. The CGM implementation for NITFS shall provide the capability to generate and output the Line Color element using the following format. Line colors are supported for use by the line primitive. RGB values are each specified using a single byte, and the last byte of the command shall be a null byte.

TABLE 93. Line Color output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
5				48							3					= 0x5083
RED								GREEN								
BLUE								0								

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5.1.2.5 Graphical Primitive elements. The Graphical Primitive elements describe the visual components of a picture in the CGM.

5.1.2.5.1 Text element output. The CGM implementation for NITFS shall provide the capability to generate and output the Text element using the following formats. The X and Y parameters give the coordinates of the lower left hand corner of the text string. The flag parameter is an integer (1) indicating that this is the final text in the string. The length parameter is an unsigned byte containing the number of characters in the string. Finally, the text is given as a character string C1, C2, ... Cn with length n. If necessary, the last byte is padded with a zero so that the next command begins on a word boundary.

TABLE 94. Text padded, short form output.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	parameter list length		
X				
Y				
1				= 0x0001
length = n		C1		
C2		. . .		
Cn		0		

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TABLE 95. Text nonpadded, short form output.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	parameter list length		
X				
Y				
1				= 0x0001
length = n (odd)		C1		
C2		. . .		
C(n-1)		Cn		

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TABLE 96. Text padded, long form output.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	31		= 0x409F
parameter list length				
X				
Y				
1				= 0x0001
length = n (even)		C1		
C2		. . .		
Cn		0		

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TABLE 97. Text nonpadded, long form output.

MSB 15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0 LSB	
4	4	31		= 0x409F
parameter list length				
X				
Y				
1				= 0x0001
length = n (odd)		C1		
C2		. . .		
C(n-1)		Cn		

5.1.2.5.2 Polygon element output. The CGM implementation for NITFS shall provide the capability to generate and output the Polygon element using the following format. The polygon parameters consist of a list of pairs of coordinates indicating the vertices of a polygon. The first vertex (Vertex1X, Vertex1Y) is connected to the last (Vertex(N)X, Vertex(N)Y) to close the polygon. Polygons are not "clipped" to the image boundary. Therefore, some coordinates may specify off-image or off-screen locations, including negative locations. Note, the parameter list length is given as the total number of bytes for all vertex parameters (4N).

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TABLE 98. Polygon short form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
4	7	parameter list length (4N)
Vertex1X		
Vertex1Y		
Vertex2X		
Vertex2Y		
. . .		
Vertex(N)X		
Vertex(N)Y		

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TABLE 99. Polygon long form output.

MSB 15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	LSB 0	
4				7							31					= 0x40FF
parameter list length (4N)																
Vertex1X																
Vertex1Y																
Vertex2X																
Vertex2Y																
.																
Vertex(N)X																
Vertex(N)Y																

5.1.2.5.3 Ellipse element output. The CGM implementation for NITFS shall provide the capability to generate and output the Ellipse element using the following format. The ellipse parameters consist of a list of three pairs of coordinates. The first pair, CenterX and CenterY, give the location of the center of the ellipse. The other two pairs, End1X, End1Y, and End2X, End2Y, specify the end points of the first and second conjugate diameters. The ellipse interior will be as specified by the Interior Style command, and the edge width, type, and color will be as specified in the Edge Width, Edge Type, and Edge Color commands respectively.

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TABLE 100. Ellipse output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
4	17	12	= 0x422C
CenterX			
CenterY			
End1X			
End1Y			
End2X			
End2Y			

5.1.2.5.4 Polyline element output. The CGM implementation for NITFS shall provide the capability to generate and output the Polyline element using the following format. The polyline parameters consist of a list of pairs of coordinates indicating the vertices of a polyline (Vertex1X, Vertex1Y) to (Vertex(N)X, Vertex(N)Y). The line width, type, and color are specified by the Line Width, Line Type, and Line Color commands, respectively. Polylines are not "clipped" to the image boundary. Therefore, some coordinates may specify off-image or off-screen locations, including negative locations. Note, the parameter list length is given as the total number of bytes for all vertex parameters (4N).

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TABLE 101. Polyline short form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB
4	1	parameter list length (4N)
Vertex1X		
Vertex1Y		
Vertex2X		
Vertex2Y		
. . .		
Vertex(N)X		
Vertex(N)Y		

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Check the source to verify that this is the current version before use.

TABLE 102. Polyline long form output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	1	31	= 0x403F
parameter list length (4N)			
Vertex1X			
Vertex1Y			
Vertex2X			
Vertex2Y			
.			
.			
.			
Vertex(N)X			
Vertex(N)Y			

5.1.2.5.5 Elliptical Arc element output. The CGM implementation for NITFS shall provide the capability to generate and output the Elliptical Arc element using the following format. A conjugate diameter pair of an ellipse is a pair of diameters of the ellipse such that a tangent to the ellipse at each endpoint is parallel to the other diameter. The centerpoint (CenterX, CenterY) specifies the center of an ellipse. The conjugate diameter endpoints (End1X, End1Y, and End2X, End2Y) include one endpoint from each conjugate diameter. Together with the centerpoint, they define the two conjugate diameters of the ellipse. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. A start ray and end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The define arc begins at the intersection of the ellipse and the end ray in the direction defined as follows. A "conjugate radius" is defined to be half of a conjugate diameter. Letting the centerpoint be labeled M, the first conjugate diameter endpoint P_1, and the second conjugate diameter endpoint P_2, then the line segments M-P_1 and M-P_2 define two conjugate radii, referred to in what follows as the first conjugate radius and the second conjugate

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radius, respectively. The conjugate radii meet at M and define two angles: the sum of the two angles is 360 degrees, one angle is less than 180 degrees and the other is greater than 180 degrees. The drawing direction of the elliptical arc is less than 180 degrees and the other is greater than 180 degrees. The drawing direction of the elliptical arc is the direction from the first conjugate radius to the second conjugate radius through the smaller of these two angles. Valid values of the three specifying points of the ellipse are those which yield three distinct points. The specified ellipse is non-degenerate if and only if the three points are non-colinear. Valid values of the vector components are those which produce vectors of non-zero length. If the start ray and end ray are coincident, it is ambiguous whether the defined arc is null (zero arc length) or the entire ellipse. The arc width, type and color are specified by the Line Width, Line Type, and Line Color commands, respectively.

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TABLE 103. Elliptical Arc output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1	LSB 0	
4	18	20	= 0x4254
CenterX			
CenterY			
End1X			
End1Y			
End2X			
End2Y			
StartVectorX			
StartVectorY			
EndVectorX			
EndVectorY			

5.1.2.5.6 Elliptical Arc Close element output. The CGM implementation for NITFS shall provide the capability to generate and output the Elliptical Arc Close element using the following format. A conjugate diameter pair of an ellipse is a pair of diameters of the ellipse such that a tangent to the ellipse at each endpoint is parallel to the other diameter. The centerpoint (CenterX, CenterY) specifies the center of an ellipse. The conjugate diameter endpoints (End1X, End1Y, and End2X, End2Y) include one endpoint from each conjugate diameter. Together with the centerpoint, they define the two conjugate diameters of the ellipse. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. A start ray and end ray are derived from the

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start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The define arc begins at the intersection of the ellipse and the end ray in the direction defined as follows. A "conjugate radius" is defined to be half of a conjugate diameter. Letting the centerpoint be labeled M, the first conjugate diameter endpoint P_1, and the second conjugate diameter endpoint P_2, then the line segments M-P_1 and M-P_2 define two conjugate radii, referred to in what follows as the first conjugate radius and the second conjugate radius, respectively. The conjugate radii meet at M and define two angles: the sum of the two angles is 360 degrees, one angle is less than 180 degrees and the other is greater than 180 degrees. The drawing direction of the elliptical arc is the direction from the first conjugate radius to the second conjugate radius through the smaller of these two angles. Valid values of the three specifying points of the ellipse are those which yield three distinct points. The specified ellipse is non-degenerate if and only if the three points are non-colinear. Valid values of the vector components are those which produce vectors of non-zero length. If the start ray and end ray are coincident, it is ambiguous whether the defined arc is null (zero arc length) or the entire ellipse. The last parameter, Close Type, specifies how the arc is closed (0=pie or 1=chord). If the close type is chord, a line is drawn between the endpoints of the arc. If the close type is pie, a line is drawn from the beginning of the arc to the centerpoint of the ellipse and then to the end of the arc. The interior of the arc will be as specified by the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

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TABLE 104. Elliptical Arc Close output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	19	22	= 0x4276
CenterX			
CenterY			
End1X			
End1Y			
End2X			
End2Y			
StartVectorX			
StartVectorY			
EndVectorX			
EndVectorY			
Close Type			

5.1.2.5.7 Rectangle element output. The CGM implementation for NITFS shall provide the capability to generate and output the Rectangle element using the following format. The rectangle is defined by two distinct points that are diagonally opposite corners of the rectangle where the rectangle is oriented parallel to the VDC axes. The interior of the rectangle will be as specified in the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

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TABLE 105. Rectangle output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	11	8	= 0x4168
Corner1X			
Corner1Y			
Corner2X			
Corner2Y			

5.1.2.5.8 Circle element output. The CGM implementation for NITFS shall provide the capability to generate and output the Circle element using the following format. The first two parameters, CenterX and CenterY, give the location of the center of the circle. Only non-negative values are valid for the radius. The interior of the circle will be as specified in the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

TABLE 106. Circle output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	12	6	= 0x4186
CenterX			
CenterY			
Radius			

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5.1.2.5.9 Circular Arc Center element output. The CGM implementation for NITFS shall provide the capability to generate and output the Circular Arc Center element using the following format. The specified radius (Radius) and centerpoint (CenterX, CenterY) define a circle. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. The start ray and end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The arc is drawn from the intersection of the circle and the start ray to the intersection of the circle and the end ray in the positive angular direction, as defined by the VDC Extent. Valid values of the vector components are those which produce distinct vectors of non-zero length. The arc width, type and color are specified by the Line Width, Line Type, and Line Color commands, respectively.

TABLE 107. Circular Arc Center output.

MSB 15 14 13 12	11 10 9 8 7 6 5	4 3 2 1 0 LSB	
4	15	14	= 0x41EE
CenterX			
CenterY			
StartX			
StartY			
EndX			
EndY			
Radius			

5.1.2.5.10 Circular Arc Center Close element output. The CGM implementation for NITFS shall provide the capability to generate and output the Circular Arc Center Close element using the following format. The specified radius (Radius) and centerpoint (CenterX, CenterY) define a circle. StartVectorX and StartVectorY define a start vector, and EndVectorX and EndVectorY define an end vector. The tails of these vectors are placed on the centerpoint. The start ray and

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end ray are derived from the start and end vectors. The start and end rays are the semi-infinite lines from the centerpoint in the direction of the start and end vectors, respectively. The arc is drawn from the intersection of the circle and the start ray to the intersection of the circle and the end ray in the positive angular direction, as defined by the VDC Extent. The last parameter, CloseType, specifies how the arc is closed (0=pie or 1=chord). If the close type is chord, a line is drawn between the endpoints of the arc. If the close type is pie, a line is drawn from the starting point through the computed arc center to the ending point. Valid values of the vector components are those which produce distinct vectors of non-zero length. The interior of this element will be as specified by the Interior Style command, and the edge width, type and color will be specified in the Edge Width, Edge Type, and Edge Color commands, respectively.

TABLE 108. Circular Arc Center Close output.

MSB 15 14 13 12	11 10 9 8 7 6 5 4 3 2 1 0 LSB	= 0x4210
4	16	16
CenterX		
CenterY		
StartX		
StartY		
EndX		
EndY		
Radius		
Close Type		

5.1.2.6 CGM binary encoding. The CGM elements are represented in the binary encoding format as described in the FIPS PUB 128 document.

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5.1.2.6.1 CGM binary encoding for output . The CGM implementation for NITFS shall provide the capability to generate and output the CGM elements in the binary encoding format.

5.2 Functional requirements. The CGM implementation for NITFS complies with the specifications established in FIPS PUB 128. The following requirements establish the element position and functional requirements for the CGM implementation for NITFS. The words "command" and "element" are used synonymously throughout this document.

5.2.1 CGM element position requirements.

5.2.1.1 CGM element position input requirements.

5.2.1.1.1 Begin Metafile element position for input. The CGM implementation for NITFS shall interpret the Begin Metafile element as the first command in the metafile.

5.2.1.1.2 Begin Picture element position for input. The CGM implementation for NITFS shall interpret the Begin Picture element after all Metafile Descriptor elements in the metafile.

5.2.1.1.3 Begin Picture Body element position for input. The CGM implementation for NITFS shall interpret the Begin Picture Body element after each Begin Picture element and its associated Picture Descriptor elements.

5.2.1.1.4 End Picture element position for input. The CGM implementation for NITFS shall interpret the End Picture element after any Begin Picture Body element and its associated Attribute and Graphical Primitive elements.

5.2.1.1.5 End Metafile element position for input. The CGM implementation for NITFS shall interpret the End Metafile element as the last element in the metafile.

5.2.1.1.6 Metafile Descriptor elements position for input. The CGM implementation for NITFS shall interpret all Metafile Descriptor elements after the Begin Metafile element and before any Begin Picture elements. The Metafile Version element, the Metafile Element List element and the Metafile Description element shall be the first three metafile descriptor elements in that order.

5.2.1.1.7 Picture Descriptor elements position for input. The CGM implementation for NITFS shall interpret all Picture Descriptor elements after the Begin Picture element and before the Begin Picture Body element.

5.2.1.1.8 Attribute elements position for input. The CGM implementation for NITFS shall interpret any Attribute Element after the Begin Picture Body element and before the Graphical Primitive element to which it applies.

5.2.1.1.9 Graphical Primitive elements position for input. The CGM implementation for NITFS shall interpret any Graphical Primitive elements after the Begin Picture Body element and any associated Attribute elements for that Graphical Primitive element and before the End Picture element.

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5.2.1.2 CGM element position output requirements.

5.2.1.2.1 Begin Metafile element position for output. The CGM implementation for NITFS shall generate the Begin Metafile element as the first command in the metafile.

5.2.1.2.2 Begin Picture element position for output. The CGM implementation for NITFS shall generate the Begin Picture element after all Metafile Descriptor elements in the metafile.

5.2.1.2.3 Begin Picture Body element position for output. The CGM implementation for NITFS shall generate the Begin Picture Body element after each Begin Picture element and its associated Picture Descriptor elements.

5.2.1.2.4 End Picture element position for output. The CGM implementation for NITFS shall generate the End Picture element after any Begin Picture Body element and its associated Attribute and Graphical Primitive elements.

5.2.1.2.5 End Metafile element position for output. The CGM implementation for NITFS shall generate the End Metafile element as the last element in the metafile.

5.2.1.2.6 Metafile Descriptor elements position for output. The CGM implementation for NITFS shall generate all Metafile Descriptor elements after the Begin Metafile element and before any Begin Picture elements. The Metafile Version element, the Metafile Element List element, and the Metafile Description element shall be the first three metafile descriptor elements in that order.

5.2.1.2.7 Picture Descriptor elements position for output. The CGM implementation for NITFS shall generate all Picture Descriptor elements after the Begin Picture element and before the Begin Picture Body element.

5.2.1.2.8 Attribute elements position for output. The CGM implementation for NITFS shall generate any Attribute Element after the Begin Picture Body element and before the Graphical Primitive element to which it applies.

5.2.1.2.9 Graphical Primitive elements position for output. The CGM implementation for NITFS shall generate any Graphical Primitive elements after the Begin Picture Body element and any associated Attribute elements for that Graphical Primitive element and before the End Picture element.

5.2.2 CGM element functional requirements.

5.2.2.1 CGM element functional input requirements.

5.2.2.1.1 CGM input required elements. The following CGM elements are required for each CGM that the CGM implementation for NITFS inputs and interprets:

a. Begin Metafile

b. Metafile Version

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c. Metafile Element List

d. Metafile Description

e. Begin Picture

f. Color Selection Mode

g. VDC Extent

h. Begin Picture Body

i. End Picture

j. End Metafile.

5.2.2.1.2 Metafile Description element contents required for input. The CGM implementation for NITFS shall be able to input and interpret the Metafile Description element that contains the following substring:

"NITF/CGM-APP-2.0."

5.2.2.1.3 Length of parameter strings required for input for the Begin Metafile, Begin Picture, and Metafile Description elements. The CGM implementation for NITFS shall be able to input and interpret the Begin Metafile, Begin Picture, and Metafile Description elements with parameter strings of at least 1024 characters.

5.2.2.1.4 Length of parameter strings required for input for the Font List element. The CGM implementation for NITFS shall be able to input and interpret the Font List element with parameter strings of at least 1024 characters.

5.2.2.1.5 Number of Begin Picture elements and Begin Picture Body elements required for input. The CGM implementation for NITFS shall be able to input and interpret one Begin Picture element with only one corresponding Begin Picture Body element.

5.2.2.1.6 End Picture element required for input. The CGM implementation for NITFS shall be able to input and interpret a CGM where an End Picture element occurs for each Begin Picture element.

5.2.2.1.7 VDC Extent element required for input. The CGM implementation for NITFS shall provide the capability to input and interpret a CGM where the VDC extent origin maps to the row and column given in the SLOC field in the National Imagery Transmission Format (NITF) Symbol Subheader and the VDC extent space maps one-to-one to the source coordinates.

5.2.2.1.8 Edge Width Specification Mode element for input. The CGM implementation for NITFS shall be able to input and interpret a CGM where the Edge Width Specification Mode element occurs before any filled-area primitive element.

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5.2.2.1.9 Line Width Specification Mode element for input. The CGM implementation for NITFS shall be able to input and interpret a CGM where the Line Width Specification Mode element occurs before any line primitive element.

5.2.2.1.10 Color Selection Mode element required for input. The CGM implementation for NITFS shall be able to input and interpret a CGM where the Color Selection Mode element occurs before any graphics primitive.

5.2.2.1.11 Character Orientation element required for input. The CGM implementation for NITFS shall be able to input and interpret a CGM where the character orientation element occurs before the Text Primitive element to ensure that the text is displayed upright and left to right.

5.2.2.1.12 Font List number required for input. The CGM implementation for NITFS shall be able to input and interpret at least 32 font name entries in the Font List element.

5.2.2.1.13 Font names for input. The CGM implementation for NITFS shall use a system font for any unsupported font name specified in the Font List element.

5.2.2.1.14 Text Font Index required for input. The CGM implementation for NITFS shall be able to input and interpret a Text Font Index element with index n when the Font List element has been interpreted with the number of fonts at least n.

5.2.2.1.15 Edge widths for input. The CGM implementation for NITFS shall be able to substitute default system edge widths for any unsupported edge widths from the Edge Width element.

5.2.2.1.16 Line widths for input. The CGM implementation for NITFS shall be able to substitute default system line widths for any unsupported line widths from the Line Width element.

5.2.2.1.17 Edge types required for input. The CGM implementation for NITFS shall be able to substitute solid edge type for any unsupported edge type from the Edge Type element.

5.2.2.1.18 Line types required for input. The CGM implementation for NITFS shall be able to substitute solid line type for any unsupported line type from the Line Type element.

5.2.2.1.19 Interior styles required for input. The CGM implementation for NITFS shall be able to substitute empty interior style for any unsupported interior style from the Interior Style element.

5.2.2.1.20 Text element requirements for input. The CGM implementation for NITFS shall be able to input and interpret the Text element with text string parameter length at least 254 characters.

5.2.2.1.21 Character Height requirements for input. The CGM implementation for NITFS shall be able to substitute default system text heights for any unsupported character height in the Character Height element.

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5.2.2.1.22 Polyline element requirements for input. The CGM implementation for NITFS shall be able to input and interpret the Polyline element with at least 4096 vertices.

5.2.2.1.23 Polygon element requirements for input. The CGM implementation for NITFS shall be able to input and interpret the Polygon element with at least 4096 vertices.

5.2.2.1.24 Input and interpret in sequential order. The CGM implementation for NITFS shall be able to interpret the CGM elements in the sequential order from which they are input from the metafile.

5.2.2.1.25 Input Text element. The CGM implementation for NITFS shall be able to input and interpret the Text element using the following attributes.

a. Text Color

b. Character Height

c. Text Font Index

d. Character Orientation.

5.2.2.1.26 Input Polygon element. The CGM implementation for NITFS shall be able to input and interpret the Polygon element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.1.27 Input Ellipse element. The CGM implementation for NITFS shall be able to input and interpret the Ellipse element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

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d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.1.28 Input Polyline element. The CGM implementation for NITFS shall be able to input and interpret the Polyline element using the following attributes.

a. Line Width

b. Line Type

c. Line Color.

5.2.2.1.29 Input Elliptical Arc element. The CGM implementation for NITFS shall be able to input and interpret the Elliptical Arc element using the following attributes.

a. Line Width

b. Line Type

c. Line Color.

5.2.2.1.30 Input Elliptical Closed Arc element. The CGM implementation for NITFS shall be able to input and interpret the Elliptical Arc Close element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.1.31 Input Rectangle element. The CGM implementation for NITFS shall be able to input and interpret the Rectangle element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

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d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.1.32 Input Circle element. The CGM implementation for NITFS shall be able to input and interpret the Circle element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.1.33 Input Circular Arc Center element. The CGM implementation for NITFS shall be able to input and interpret the Circular Arc Center element using the following attributes.

a. Line Width

b. Line Type

c. Line Color.

5.2.2.1.34 Input Circular Arc Center Close element. The CGM implementation for NITFS shall be able to input and interpret the Circular Arc Center Close element using the following attributes.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

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5.2.2.1.35 CGM element defaults for input. The CGM implementation for NITFS shall assume all CGM default values as stated in the Part 3 - Binary Encoding in the FIPS PUB 128 for each CGM that the CGM implementation inputs and interprets unless otherwise specified in this document. This table includes but is not limited to the following:

TABLE 109. CGM element defaults for input.

VDC TYPE:	16 BIT INTEGER
INTEGER PRECISION:	16 BIT INTEGER
INDEX PRECISION:	16 BIT INTEGER
COLOR PRECISION:	8 BIT INTEGER
TRANSPARENCY:	ON
LINE TYPE:	1 (SOLID)
TEXT PRECISION:	STRING
CHARACTER EXPANSION FACTOR:	1.0
CHARACTER SPACING:	0.0
CHARACTER ORIENTATION:	0, 1, 1, 0
TEXT PATH:	RIGHT
TEXT ALIGNMENT:	NORMAL HORIZONTAL, NORMAL VERTICAL
INTERIOR STYLE:	HOLLOW (EMPTY)
EDGE TYPE:	SOLID
EDGE VISIBILITY:	OFF
LINE COLOR:	DEVICE-DEPENDENT FOREGROUND COLOR
EDGE COLOR:	DEVICE-DEPENDENT FOREGROUND COLOR
FILL COLOR:	DEVICE-DEPENDENT FOREGROUND COLOR
TEXT COLOR:	DEVICE-DEPENDENT FOREGROUND COLOR
BACKGROUND COLOR:	NONE (THIS IS NITF SPECIFIC)
COLOR VALUE EXTENT:	0,0,0 - 255,255,255
VDC INTEGER PRECISION:	16 BIT INTEGER
TEXT FONT INDEX:	1

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5.2.2.1.36 Default colors for unsupported text. The CGM implementation for NITFS shall substitute available system colors for unsupported colors specified in the Text Color element, Fill Color element, Edge Color element, and the Line Color element.

5.2.2.1.37 CGM element substitution. The CGM implementation for NITFS shall report or substitute for any CGM element and associated parameters not supported in the CGM implementation for NITFS and continue to interpret the next element supported in the CGM implementation for the NITFS.

5.2.2.1.38 CGM error messages. The CGM implementation for NITFS should report errors encountered during the input and interpretation of the CGM.

5.2.2.2 CGM element functional output requirements.

5.2.2.2.1 CGM output required elements. The following CGM elements are required for each CGM that the CGM implementation for NITFS generates and outputs:

a. Begin Metafile

b. Metafile Version

c. Metafile Element List

d. Metafile Description

e. Begin Picture

f. Color Selection Mode

g. VDC Extent

h. Begin Picture Body

i. End Picture

j. End Metafile.

5.2.2.2.2 Metafile Description element contents required for output. The CGM implementation for NITFS shall generate and output the Metafile Description element that contains the following substring:

"NITF/CGM-APP-2.0."

5.2.2.2.3 Length of parameter strings required for output for the Begin Metafile, Begin Picture, and Metafile Description elements. The CGM implementation for NITFS shall generate and output the Begin Metafile, Begin Picture, and Metafile Description elements with parameter strings not to exceed 1024 characters.

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5.2.2.2.4 Length of parameter strings required for output for the Font List element. The CGM implementation for NITFS shall generate and output the Font List element with parameter strings not to exceed 1024 characters.

5.2.2.2.5 Number of Begin Picture elements and Begin Picture Body elements required for output. The CGM implementation for NITFS shall generate and output only one Begin Picture element with only on Begin Picture Body element.

5.2.2.2.6 End Picture element required for output. Exactly one End Picture element is required for each Begin Picture element for each CGM that the CGM implementation for NITFS generates and outputs.

5.2.2.2.7 Edge Width Specification Mode element for output. The Edge Width Specification Mode element is required when any filled-area primitive element is present for each CGM that the CGM implementation for NITFS generates and outputs.

5.2.2.2.8 Line Width Specification Mode element for output. The Line Width Specification Mode element is required when any line primitive element is present for each CGM that the CGM implementation for NITFS generates and outputs.

5.2.2.2.9 Color Selection Mode element required for output. The Color Selection Mode element is required for each CGM that the CGM implementation for NITFS generates and outputs.

5.2.2.2.10 Character Orientation element required for output. The character orientation element is required for the Text Primitive element when the VDC Extent element parameters are given as X1 > X2 or Y1 > Y2. This ensures that the text is displayed upright and left to right for each CGM that the CGM implementation for NITFS generates and outputs.

5.2.2.2.11 Font List number required for output. The CGM implementation for NITFS shall provide the capability to generate and output, at most, 32 font name entries in the Font List element.

5.2.2.2.12 Font names for output. The CGM implementation for NITFS shall limit the font name in the Font List element to the following list. The HERSHEY fonts are specified in NBS SP 424 while TIMES, COURIER and HELVETICA fonts are registered trademarks of Allied Corporation which owns the copyright.

a. HERSHEY/CARTOGRAPHIC_ROMAN

b. HERSHEY/CARTOGRAPHIC_GREEK

c. HERSHEY/SIMPLEX_ROMAN

d. HERSHEY/SIMPLEX_GREEK

e. HERSHEY/SIMPLEX_SCRIPT

f. HERSHEY/COMPLEX_ROMAN

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g. HERSHEY/COMPLEX_GREEK

h. HERSHEY/COMPLEX_SCRIPT

i. HERSHEY/COMPLEX_ITALIC

j. HERSHEY/COMPLEX_CYRILLIC

k. HERSHEY/DUPLEX_ROMAN

l. HERSHEY/TRIPLEX_ROMAN

m. HERSHEY/TRIPLEX_ITALIC

n. HERSHEY/GOTHIC_GERMAN

o. HERSHEY/GOTHIC_ENGLISH

p. HERSHEY/GOTHIC_ITALIAN

q. TIMES_ROMAN

r. TIMES_ITALIC

s. TIMES_BOLD

t. TIMES_BOLD_ITALIC

u. HELVETICA

v. HELVETICA_OBLIQUE

w. HELVETICA_BOLD

x. HELVETICA_BOLD_OBLIQUE

y. COURIER

z. COURIER_BOLD

aa. COURIER_ITALIC

ab. COURIER_BOLD_ITALIC

5.2.2.2.13 Text Font Index required for output. The index of every Text Font Index element shall be less than or equal to (≤) the number of fonts specified in the Font List element for any CGM implementation generated for NITFS. The Font List element is required for each CGM that contains a Text Font Index element.

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5.2.2.2.14 Edge widths for output. The CGM implementation for NITFS shall provide the capability to generate and output edge widths of two, four, or six for the Edge Width element.

5.2.2.2.15 Line widths for output. The CGM implementation for NITFS shall provide the capability to generate and output line widths of two, four, or six for the Line Width element.

5.2.2.2.16 Edge types required for output. The CGM implementation for NITFS shall generate and output edge types of solid (1) and dashed (2) to the Edge Type element.

5.2.2.2.17 Line types required for output. The CGM implementation for NITFS shall generate and output line types of solid (1) and dashed (2) to the Line Type element.

5.2.2.2.18 Interior styles required for output. The CGM implementation for NITFS shall generate and output interior styles of solid (1) and empty (4) to the Interior Style element.

5.2.2.2.19 Text element requirements for output. The CGM implementation for NITFS shall generate and output the Text element with text string parameter not to exceed 254 characters.

5.2.2.2.20 Character Height requirements for output. The CGM implementation for NITFS shall generate and output the Character Height element within 10 through 38 inclusive.

5.2.2.2.21 Polyline element requirements for output. The CGM implementation for NITFS shall generate and output the Polyline element with at least two and no more than 4096 vertices.

5.2.2.2.22 Polygon element requirements for output. The CGM implementation for NITFS shall generate and output the Polygon element with at least three vertices and no more than 4096 vertices.

5.2.2.2.23 Generate and output sequential order. The CGM implementation for NITFS shall generate and output CGM elements in the sequential order intended for interpretation.

5.2.2.2.24 Output Text element. The CGM implementation for NITFS shall generate and output the Text element using the following attributes. The following attributes must be stated before the first use of the Text element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Text Color

b. Character Height

c. Text Font Index

d. Character Orientation.

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5.2.2.2.25 Output Polygon element. The CGM implementation for NITFS shall generate and output the Polygon element using the following attributes. The following attributes must be stated before the first use of the Polygon element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.2.26 Output Ellipse element. The CGM implementation for NITFS shall generate and output the Ellipse element using the following attributes. The following attributes must be stated before the first use of the Ellipse element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.2.27 Output Polyline element. The CGM implementation for NITFS shall generate and output the Polyline element using the following attributes. The following attributes must be stated before the first use of the Polyline element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Line Width

b. Line Type

c. Line Color.

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5.2.2.2.28 Output Elliptical Arc element. The CGM implementation for NITFS shall generate and output the Elliptical Arc element using the following attributes. The following attributes must be stated before the first use of the Elliptical Arc element but need not be restated for subsequent occurrence unless a different attribute value is required.

a. Line Width

b. Line Type

c. Line Color.

5.2.2.2.29 Output Elliptical Arc Close element. The CGM implementation for NITFS shall generate and output the Elliptical Arc Close element using the following attributes. The following attributes must be stated before the first use of the Elliptical Arc Close element but need not be restated for subsequent occurrence unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.2.30 Output Rectangle element. The CGM implementation for NITFS shall generate and output the Rectangle element using the following attributes. The following attributes must be stated before the first use of the Rectangle element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

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5.2.2.2.31 Output Circle element. The CGM implementation for NITFS shall generate and output the Circle element using the following attributes. The following attributes must be stated before the first use of the Circle element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.2.32 Output Circular Arc Center element. The CGM implementation for NITFS shall generate and output the Circular Arc Center element using the following attributes. The following attributes must be stated before the first use of the Circular Arc Center element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Line Width

b. Line Type

c. Line Color.

5.2.2.2.33 Output Circular Arc Center Close element. The CGM implementation for NITFS shall generate and output the Circular Arc Center Close element using the following attributes. The following attributes must be stated before the first use of the Circular Arc Center Close element but need not be restated for subsequent occurrences unless a different attribute value is required.

a. Fill Color

b. Interior Style

c. Edge Visibility

d. Edge Width

e. Edge Type

f. Edge Color.

5.2.2.2.34 CGM element defaults for output. The CGM implementation for NITFS shall assume all CGM default values as stated in the Part 3 - Binary Encoding in the FIPS PUB 128 for

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each CGM that the CGM implementation generates and outputs unless otherwise specified in this document. This table includes but is not limited to the following:

TABLE 110. CGM element defaults for output.

VDC TYPE:	16 BIT INTEGER
INTEGER PRECISION:	16 BIT INTEGER
INDEX PRECISION:	16 BIT INTEGER
COLOR PRECISION:	8 BIT INTEGER
TRANSPARENCY:	ON
TEXT PRECISION:	STRING
CHARACTER EXPANSION FACTOR:	1.0
CHARACTER SPACING:	0.0
TEXT PATH:	RIGHT
TEXT ALIGNMENT:	NORMAL HORIZONTAL, NORMAL VERTICAL
BACKGROUND COLOR:	NONE (THIS IS NITF SPECIFIC)
COLOR VALUE EXTENT:	0,0,0 - 255,255,255
VDC INTEGER PRECISION:	16 BIT INTEGER

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6. NOTES

(This section contains general or explanatory information that may be helpful but is not mandatory).

6.1 CGM example metafiles. The following examples illustrate complete CGM metafiles compatible with the CGM implementation for NITFS.

6.1.1 Polygon example. The following example represents a white filled polygon with three vertices at (4,5), (6,7), and (8,9) relative to the VDC Extent origin.

TABLE 111. Polygon example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("POLYGON")	0x0028 0x0750 0x4F4C 0x5947 0x4F4E
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("POLYGON")	0x1048 0x0750 0x4F4C 0x5947 0x4F4E
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
BEGIN PICTURE ("POLYGON")	0x0068 0x0750 0x4F4C 0x5947 0x4F4E
COLOR SELECTION MODES (DIRECT = 1)	0x2042 0x0001

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TABLE 111. Polygon example - Continued.

CGM ELEMENT	HEX VALUES
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000
BEGIN PICTURE BODY	0x0080
FILL COLOR (255,255,255)	0x52E3 0xFFFF 0xFF00
INTERIOR STYLE (SOLID=1)	0x52C2 0x0001
POLYGON (4,5 6,7 8,9)	0x40FF 0x000C 0x0004 0x0005 0x0006 0x0007 0x0008 0x0009
END PICTURE	0x00A0
END METAFILE	0x0040

6.1.2 Text example. The following CGM example represents a label with text "test" with white characters 14 pixels high starting at location (10,20) relative to VDC Extent origin. the font list consists of two fonts (Helvetica and Courier). The text uses Courier.

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TABLE 112. Text example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("TEXT")	0x0025 0x0454 0x4558 0x5400
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("TEXT")	0x1045 0x0454 0x4558 0x5400
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
FONT LIST ("Helvetica," "Courier")	0x11B2 0x0948 0x656C 0x7665 0x7469 0x6361 0x0743 0x6F75 0x7269 0x6572
BEGIN PICTURE ("TEXT")	0x0065 0x0454 0x4558 0x5400
COLOR SELECTION MODES (DIRECT = 1)	0x2042 0x0001
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000

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TABLE 112. Text example - Continued.

CGM ELEMENT	HEX VALUES
BEGIN PICTURE BODY	0x0080
TEXT COLOR (255,255,255)	0x51C3 0xFFFF 0xFF00
CHARACTER HEIGHT (14)	0x51E2 0x000E
TEXT FONT INDEX (2=Courier)	0x5142 0x0002
CHARACTER ORIENTATION	0x5208 0x0000 0xFFFF 0x0001 0x0000
TEXT (10,20,1,4,"test")	0x408B 0x000A 0x0014 0x0001 0x0474 0x6573 0x7400
END PICTURE	0x00A0
END METAFILE	0x0040

6.1.3 Ellipse example. The following CGM example represents an empty ellipse with a four-pixel wide visible white edge, a center at location (10,20), with conjugate endpoints at locations (20,20) and (10,30) relative to VDC Extent origin.

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TABLE 113. Ellipse example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("ELLIPSE")	0x0028 0x0745 0x4C4C 0x4950 0x5345
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("ELLIPSE")	0x1048 0x0745 0x4C4C 0x4950 0x5345
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
BEGIN PICTURE ("ELLIPSE")	0x0068 0x0745 0x4C4C 0x4950 0x5345
COLOR SELECTION MODES (DIRECT = 1)	0x2042 0x0001
EDGES WIDTH SPECIFICATION MODE (ABSOLUTE=0)	0x20A2 0x0000
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000
BEGIN PICTURE BODY	0x0080

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TABLE 113. Ellipse example - Continued.

CGM ELEMENT	HEX VALUES
EDGE COLOR (255,255,255)	0x52A3 0xFFFF 0xFF00
EDGE VISIBILITY (on)	0x53C2 0x0001
EDGE WIDTH (4)	0x5382 0x0004
EDGE TYPE (1=solid)	0x5362 0x0001
INTERIOR STYLE (EMPTY)	0x52C2 0x0004
ELLIPSE (10,20 20,20 10,30)	0x422C 0x000A 0x0014 0x0014 0x0014 0x000A 0x001E
END PICTURE	0x00A0
END METAFILE	0x0040

6.1.4 Polyline example. The following CGM example represents a three-segment open RED polyline drawn with a six-pixel wide dashed line. The vertices are at (2,2), (4,6), (3,4), and (0,6) relative to VDC Extent origin.

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TABLE 114. Polyline example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("POLYLINE")	0x0029 0x0850 0x4F4C 0x594C 0x494E 0x4500
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("POLYLINE")	0x1049 0x0850 0x4F4C 0x594C 0x494E 0x4500
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
BEGIN PICTURE ("POLYLINE")	0x0069 0x0850 0x4F4C 0x594C 0x494E 0x4500
COLOR SELECTION MODE (DIRECT = 1)	0x2042 0x0001
LINE WIDTH SPECIFICATION MODE (ABSOLUTE=0)	0x2062 0x0000
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000

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TABLE 114. Polyline example - Continued.

CGM ELEMENT	HEX VALUES
BEGIN PICTURE BODY	0x0080
LINE COLOR (255,0,0)	0x5083 0xFF00 0x0000
LINE TYPE (DASH=2)	0x5042 0x0002
LINE WIDTH (6)	0x5062 0x0006
POLYLINE (2,2 4,6 3,4 0,6)	0x403F 0x0010 0x0002 0x0002 0x0004 0x0006 0x0003 0x0004 0x0000 0x0006
END PICTURE	0x00A0
END METAFILE	0x0040

6.1.5 Elliptical Arc example. The following CGM example represents a blue elliptical arc that is four pixels wide. Figure 6 demonstrates pictorially the Elliptical Arc parameters and the relationship between conjugate diameter endpoints and start and stop vectors.

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TABLE 115. Arc example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("ARC")	0x0024 0x0341 0x5243
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("ARC")	0x1048 0x0341 0x5243
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
BEGIN PICTURE ("ARC")	0x0068 0x0341 0x5243
COLOR SELECTION MODES (DIRECT = 1)	0x2042 0x0001
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000
BEGIN PICTURE BODY	0x0080
LINE WIDTH (4)	0x5062 0x0006
LINE COLOR (0,0,255)	0x5083 0x0000 0xFF00

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TABLE 115. Arc example - Continued.

CGM ELEMENT	HEX VALUES
LINE TYPE (1=solid)	0x5042 0x0001
ELLIPTICAL ARC (10,10 10,5 20,10, 15,5 5,5)	0x4254 0x0014 0x000A 0x000A 0x000A 0x0005 0x0014 0x000A 0x000F 0x0005 0x0005 0x0005
END PICTURE	0x00A0
END METAFILE	0x0040

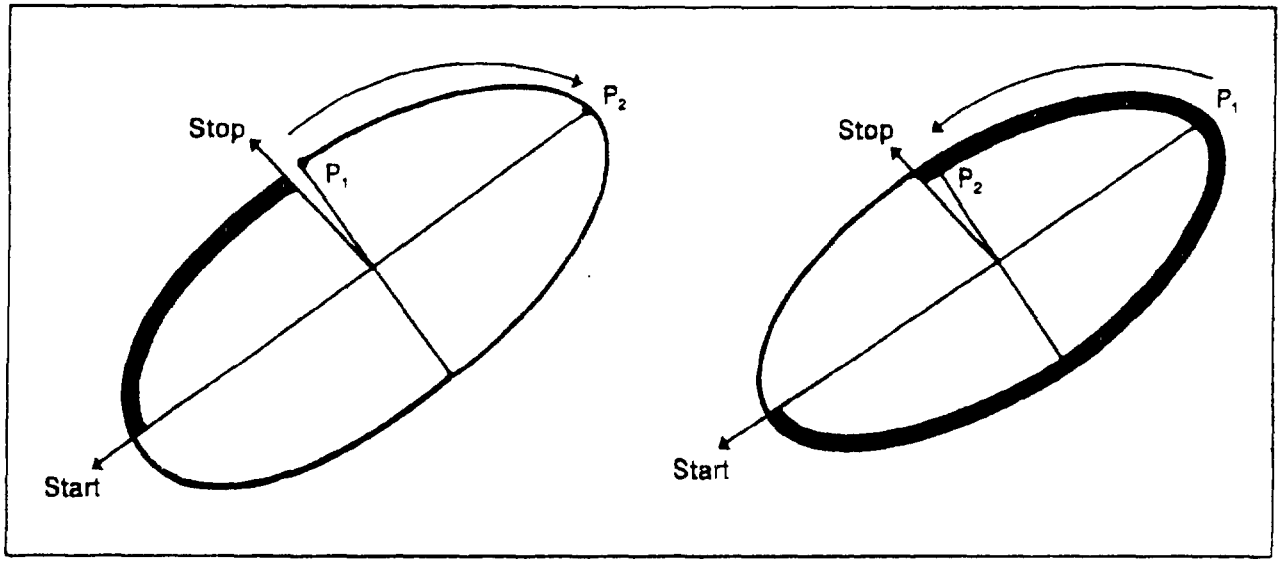


FIGURE 6. Elliptical Arc orientation.

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6.1.6 Elliptical Arc example. The following CGM example represents an elliptical arc close that is red and filled with pie closure.

TABLE 116. Arc Closed example.

CGM ELEMENT	HEX VALUES
BEGIN METAFILE ("ARCCLOSE")	0x0024 0x0841 0x5243 0x434C 0x4F53 0x4500
METAFILE VERSION (1)	0x1022 0x0001
METAFILE DESCRIPTION ("ARC")	0x1048 0x0841 0x5243 0x434C 0x4F53 0x4500
METAFILE ELEMENT LIST	0x1166 0x0001 0xFFFF 0x0001
BEGIN PICTURE ("ARC")	0x0068 0x0841 0x5243 0x434C 0x4F53 0x4500
COLOR SELECTION MODE (DIRECT = 1)	0x2042 0x0001
VDC EXTENT	0x20C8 0x0000 0x7FFF 0x7FFF 0x0000
BEGIN PICTURE BODY	0x0080

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TABLE 116. Arc Closed example - Continued.

CGM ELEMENT	HEX VALUES
INTERIOR STYLE	0x5062 0x0006
FILL COLOR (255,0,0)	0x5083 0x0000 0xFF00
EDGE VISIBILITY (on)	0x53C2 0x0001
ELLIPTICAL ARC CLOSE (10,10 10,5 20,10, 15,5 5,5)	0x4276 0x000A 0x000A 0x000A 0x0005 0x0014 0x000A 0x000F 0x0005 0x0005 0x0005 0x0000
END PICTURE	0x00A0
END METAFILE	0x0040

6.2 Color to gray scale conversion. Full color may be specified for Attribute Elements dealing with color. Color items for receiving systems unable to support full color must be mapped to colors they are able to support.

For eight-bit gray scale systems:

pixel $value_8$ = 0.299 * RED + 0.587 * GREEN + 0.114 * BLUE

For one bit black and white systems:

pixel $value_1$ = 1 (white), if pixel $value_8 > 127$

0 (black), if pixel $value_8 \leq 127$

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6.3 Subject term (key word) listing.

Image
Picture
Secondary Imagery Dissemination System
SIDS
Symbol(s)

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CONCLUDING MATERIAL

Custodians:
Army - SC
Navy - OM
Air Force - 02
Misc - DC

Preparing activity:
Misc - DC

Agent:
Not applicable

Review activities:
OASD - SO, DO, HP, IR
Army - AM, AR, MI, TM, MD, CE, SC, IE, ET, AC, PT
Navy - OM
Air Force - 02
DLA - DH
Misc - NS, MP, DI, NA

(Project IPSC-0321)

Civil agency coordinating activities:
USDA - AFS, APS
COM - NIST
DOE
EPA
GPO
HHS - NIH
DOI - BLM, GES, MIN
DOT - CGCT

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-2301	2. DOCUMENT DATE (YYMMDD) 930618

3. DOCUMENT TITLE: COMPUTER GRAPHICS METAFILE (CGM) IMPLEMENTATION STANDARD FOR THE NITFS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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c. ADDRESS (Include Zip Code) 11440 ISAAC NEWTON SQUARE, NORTH RESTON, VA 22090-5006	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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